NELSON MANDELA UNIVERSITY

Mobile Assessment and Annotation Tool

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DECLARATION

I, Joshua Page, hereby declare that the treatise for the degree Bachelor of Science Honours in Computer Science and Information Systems is my own work and that it has not previously been submitted for assessment of completion of any postgraduate qualification to another university or for another qualification.

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SUMMARY

Academic assessment has quickly adapted to digital means of evaluation by providing online assessment submissions. Despite this provision for online submissions, academic assessment has yet to progress beyond traditional marking techniques. This research aimed to develop a functional prototype of a Mobile Assessment and Annotation Tool to provide the same benefits and functionality as traditional marking techniques while eliminating drawbacks, such as possible human error and the additional administrative burden associated with manual processes.

The Design Science Research (DSR) methodology was implemented throughout research, design, development, and evaluation to deliver this prototype. Thorough research was done to establish the problem and to fully comprehend the problem this project aims to solve. Requirements were collected by evaluating extant systems and issuing questionnaires to lecturers. These requirements formed the basis of the prototype's design, which includes the initial use case activity diagram, database design, and user interfaces aimed at enhancing the electronic assessment process for both students and educators.

Implementation followed the initial prototype design as closely as possible, with inevitable alterations made to accommodate practical constraints and user feedback. After the prototype implementation was completed, evaluation took place through formative evaluation and usability testing, identifying the strengths, weaknesses, and areas that can be improved. Post evaluation, various suggestions regarding functionality and user experience were implemented, such as simplifying confusing terminology and improving the user interface, further strengthening the prototype's usability and effectiveness. A final retrospective highlighted the challenges faced during the project, insights gained, and suggestions for future work, provided this research project is extended in the future.

GLOSSARY

Term	Definition
4IR	Fourth Industrial Revolution
CLI	Command Line Interface
CNN	Convolutional Neural Network
HEI	Higher Education Institution
ICT	Information and Communication Technology
IDE	Interactive Developer Environment
MAAT	Mobile Assessment and Annotation Tool
PDF	Portable Document Format
Pedagogical	Related to teaching in some manner
SUS	System Usability Scale
UML	Unified Modelling Language

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1 INTRODUCTION

The Fourth Industrial Revolution (4IR), beginning in 2011, has ushered in a new digital age. Several technological advancements have led to the digitalisation of various sectors, fundamentally changing several industries (Ross & Maynard, 2021). This paradigm shift has led to a remarkable transformation in how various organisations operate, requiring a digitally proficient workforce to compete and excel (Malhotra, 2023).

Due to the technological advancements of the 4IR, the demand for various occupations in technological sectors has increased. For example, there are projected to be 28.7 million software developers in 2024, which is a significant increase compared to 13 million developers in 2007 (Morgan, 2007; Vailshery, 2023).

As a result of the required digital proficiency brought by the 4IR, educational institutions offering programs that teach students the required digital proficiency skills have increased their intake of students to accommodate this demand. For example, Nelson Mandela University's first-year Computer Science programming module, Programming Fundamentals 1.1, had roughly 200 students in 2019 and now has approximately 500 students in 2024.

An increase in registered students implies an increase in submissions to assess. Submissions are often submitted electronically, leading to the following choices in marking these submissions: printing and assessing a physical copy of the submission or electronically assessing a submission.

Consequently, in modules where submissions are assessed electronically, the process of marking submissions has posed significant challenges. Although traditional assessment has advantages, it also possesses several limitations, creating an inefficient and tedious assessment process. These traditional marking techniques fail to align with the technological advancements encouraged by 4IR, thereby restricting pedagogical innovation.

Recognising the need for a more effective and streamlined approach to electronic assessment, this treatise aims to explore and evaluate the feasibility of implementing a Mobile Assessment and Annotation Tool (MAAT). This project aims to develop an application for an Android Tablet that allows the assessor to mark electronic submissions in a free format.

1.1 Problem Background

Utilising traditional marking techniques for assessing electronic assessments imposes several limitations, such as human error and increased time consumption. These techniques often involve printing out each electronic submission to assess in a physical format, manually calculating submission marks, and manual capturing of each submission mark. The increased time required to manually perform these tasks, combined with the potential for errors during the summation or recording of marks introduce significant risks.

Additionally, these techniques lack the necessary flexibility and functionality, meaning they do not provide features such as automated marking symbol recognition or publishing results to mark electronic assessments promptly and efficiently. Marking submissions manually introduces numerous risks, primarily due to the possibility of human error. Such risks may have dire consequences, such as failing a student on a final exam due to an incorrect summation of all their marks achieved or incorrectly recording a student's mark.

Assessments are an integral part of the learning process; however, to truly maximise their educational value, assessors should provide relevant feedback by annotating submissions. The repetitive nature of writing and rewriting the same annotation regarding a standard error across multiple scripts can become tedious and frustrating. This cumbersome experience can encourage the lecturer to stop providing this annotation to scripts entirely, thus reducing any explanation of a mistake to students when viewing their feedback.

A significant limitation of traditional assessment is the use of printed resources. Many printed resources, such as paper and printer cartridge ink, are used when manually marking and annotating test scripts, which not only incur higher costs but also contribute to environmental degradation.

However, traditional marking techniques still have their benefits, the core benefit being the ability to mark in a free format. Free format marking enables the assessor to assess and annotate in any format which suits them, not constraining them to any specific convention. Annotations in a free format allow assessors to express their initial instincts and impressions when marking instead of focusing on complying with a predefined format.

1.2 Problem Description

Present marking techniques impose several limitations on the assessment process, with this treatise's specific focus on assessing electronic submissions submitted in Portable Document Format (PDF). There exists a need for a tool to assess digital submissions in a free format, integrating current marking techniques and their advantages, creating a more efficient and streamlined marking process for assessors.

1.3 Project Aims

This project aims to develop a functioning tool for Mobile Assessment and Annotation, providing a similar experience to traditional marking techniques with as few of the shortcomings they entail as possible. Since the natural marking process varies by person, this tool cannot be person or process-specific; it must be generalised to function for multiple types of users. This tool must provide flexibility similar to traditional assessment while enhancing the assessment process. Some of the enhancements include automatic summation of marks in a submission, automatic recording of these marks in a useful and legible electronic format, and automatic publishing of annotated scripts. These enhancements aim to culminate in an improved marking experience for assessors.

1.4 Functionality

As outlined in the project proposal (see *Appendix A: Project Proposal*), the expected functionality of the tool is as follows:

- Load all submitted tasks for a particular assessment into the system's database;
- Support for the use of a digital tablet to mark (with a special symbol, e.g., ✓) the tasks and add annotations to the submitted tasks in free format;
- Save the annotated tasks for later reference by the moderator(s) and relevant student;
- Automatically recognise special symbols (e.g., ✓) and perform an appropriate action (e.g., in the case of ✓, accumulate a count of them and use this mark as a mark awarded for the assessment);

- Store the mark awarded for a particular assessment and support the exporting of this data in a format that is compatible with an Excel spreadsheet;
- On request from the assessor, email selected (or all) assessments to the moderator and
- On request from the assessor, email each student his/her annotated assessment.

The functionality of the tool will at least need to exhibit the above-listed functionality, and future requirements for the tool will be identified in Chapter 3.

1.5 Research Methodology

This project will follow the Design Science Research (DSR) process model, beginning with research to become fully aware of the problem. Subsequently, potential solutions will be suggested based on current knowledge at the current iteration. Following iterations will refine solutions, incorporate feedback, and make necessary adjustments along the way. Continuous evaluation ensures that solutions are thoroughly assessed for reliability and effectiveness. Finally, solutions are validated to ensure that they address the identified problem as part of the conclusion (van der Merwe et al., 2020). Figure 1-1 describes the knowledge flows, process steps and outputs of the DSR methodology.

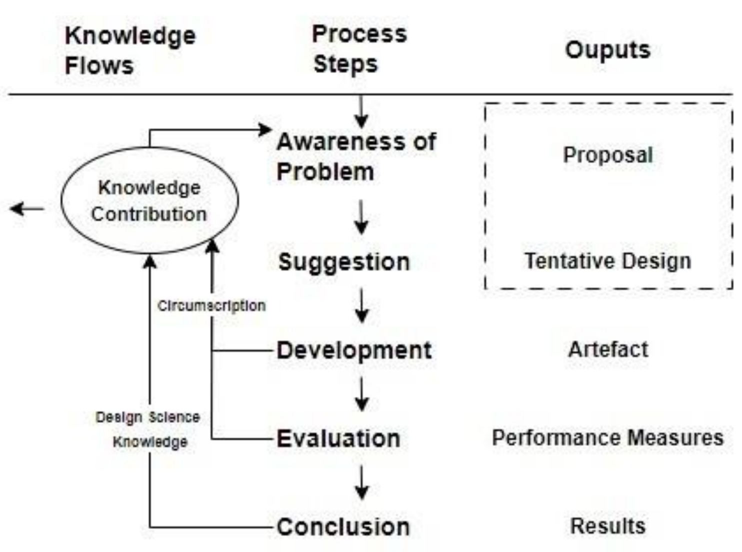

Figure 1-1: DSR Process

1.6 Contribution to the domain of assessment

The research aims to contribute to electronic assessment by developing a solution that possesses the advantages of traditional assessment while eliminating the drawbacks. This approach aims to streamline the assessment process, addressing inefficiencies and creating a smoother process.

1.7 Overview of Treatise Structure

The introductory chapter outlines the problem background while defining a few functional requirements that must be satisfied, which is further expanded on in Section 3.3. The second chapter, Literature Review, discusses the appropriate literature relevant to the domain of electronic assessment, considering extant systems and their advantages and disadvantages.

The third chapter discusses the design of the prototype, considering the functionality from extant systems in Chapter 2, as well as the identified functional requirements which inform and justify the remaining design choices in that chapter. The fourth chapter discusses the implementation of the proposed system, as well as any changes that have occurred and why.

The fifth chapter will evaluate the prototype through formative evaluation and usability testing. The evaluation will highlight weak points and areas of improvement, which will be implemented to improve the system further. The last chapter will discuss the overall journey, the various challenges and insights the researcher gained, and highlighting areas in which the prototype can further be expanded and improved upon.

2 LITERATURE REVIEW

2.1 Introduction

This chapter aims to continue with the first DSR process step, *Awareness of Problem*. The research discussed in this chapter aims to provide sufficient problem awareness, which can thus be used in the second process step of the DSR methodology, *Suggestion*, which is addressed in Chapter 3.

This chapter includes an in-depth discussion of electronic assessment, commonly referred to as e-assessment, detailing its various implications, advantages, and associated disadvantages. Traditional and electronic assessment is compared, and the importance of feedback, which both electronic and traditional assessment offer, is highlighted. Furthermore, the shift to electronic assessment, discussed in Section 2.2, further argues that a tool focused on assessing electronic assessments can forgo the limitations imposed by traditional assessment.

Various extant systems for e-assessment are also evaluated, acknowledging the areas in which they excel and are flawed, with a specific focus on the aspects that allow for manual marking and annotations. The findings from this evaluation form the basis of functional requirements discussed in Chapter 3.

2.2 Electronic Assessment

Electronic assessment is defined as any method utilising information technology to conduct any assessment-related activities (Brink & Lautenbach, 2011). E-assessment has become widely implemented at Higher Education Institutions (HEIs) because it can streamline administrative processes, create a more efficient assessment process, and provide timely and detailed feedback to students (Al-Fraihat et al., 2020).

E-assessment is typically conducted through a Learning Management System (LMS), such as Moodle or Blackboard, which have been implemented at several higher education institutions. Dahlstrom et al. (2014) reported that 99% of the institutions surveyed had implemented an LMS, and 83% of students used an LMS in some manner.

The COVID-19 pandemic in early 2020 necessitated a rapid shift to e-learning and remote work, leading to a significant increase in the adoption of e-assessment. Several lockdowns and the necessity to continue with educational activities only encouraged this increase in e-assessment and e-learning, in which using an LMS was critical. Although LMSs were utilised before 2020, the pandemic further strengthened their use in most higher education institutions, which has not changed since the return to face-to-face educational activities (Badaru & Adu, 2022).

It is worth noting that traditional and electronic assessment are not necessarily distinct concepts; rather, e-assessment is a necessary adaption to the increase of Information and Communications Technology (ICT) facilitated by the 4IR in educational assessment (Alruwais et al., 2018). According to Kiryakova (2021), e-assessment offers new options and methodologies for evaluating different levels and elements of knowledge, skills, and competencies.

2.2.1 Advantages of Electronic Assessment

Kiryakova (2021) highlighted some of the advantages of electronic assessment in various contexts. In the context of simple assessments such as multiple-choice quizzes, E-assessment allows for immediate and personalised feedback, which can guide students due to the possibility of automation. E-assessment also offers a more flexible approach, allowing various assessment techniques, such as multiple-choice questions or free-response questions, to be applied depending on the pedagogical strategy.

From the viewpoints of lecturers, Waheeda et al. (2023) examined the benefits of online assessment, a form of e-assessment. Lecturers reported an increased efficiency in assessing online submissions compared to traditional submissions and an easier method to analyse and evaluate collective and individual assessment performance. Furthermore, lecturers reported the increased ease of providing student feedback, which was more efficient through e-assessment.

Huda & Siddiq (2020) surveyed e-assessment from the perspectives of students. The survey outcome demonstrated that students were positive about e-assessments, particularly considering the reliability of assessments that contained questions in which answers can be assessed automatically, such as multiple-choice questions.

Alruwais et al. (2018) outlined how e-assessment reduces the workload of lecturers, such as reducing the amount of time required to assess submissions, thus improving the efficiency of the assessment process. This also streamlines the overall administration process, as tedious tasks such as grading and recording can be automated. Additionally, e-assessment does not require physical storage space or printing out electronic submissions to assess, thus reducing the physical and financial costs that could be incurred.

2.2.2 Disadvantages of Electronic Assessment

(Alruwais et al., 2018) investigated various challenges and disadvantages of e-assessment. E-assessment requires digital proficiency from students and assessors to engage with the assessment process successfully. E-assessment also requires an electronic device and, most often, internet access.

Although HEIs often have this required equipment, students in their personal homes may not. Additionally, embracing e-assessment may prove difficult for students and lecturers already accustomed to traditional assessment means, which may prove difficult to transition fully to e-assessment if students or lecturers resist this change.

Lastly, e-assessment can reduce some costs, but it may introduce entirely new costs associated with the required technological infrastructure and technical support. These financial demands can be a barrier to fully implementing e-assessment, especially for institutions with many students and a limited budget.

2.3 Comparison between Electronic and Traditional Assessment

Table 2-1 provides a comparative overview of electronic and traditional assessments across various aspects. This comparison highlights each assessment method's distinct advantages and disadvantages, facilitating a clearer understanding of their impacts and which aspects each method excels or falls short of.

Table 2-1: Comparing electronic and traditional assessments per aspect

Aspect	**E-assessment**	**Traditional assessment**
Feedback	Feedback is typically provided quicker than traditional assessment.	Requires manual marking, which can lead to delayed feedback.
Cost	Lower printing and storage costs, high initial setup, and possible maintenance costs could be barriers to entry.	Higher costs for printing, storage, and stationery.
Environmental impact	Reduced paper usage and lower environmental impact.	Increased paper usage and higher environmental impact.
Reliability	Dependant on technology, server status, and internet connection.	Does not require any peripherals or servers, only physical submissions and thus is more stable.
Administrative burden	Decrease administrative burden; many tasks can be automated and done efficiently.	Increased administrative burden since most tasks need to be done manually.
Process complexity	Involves a more complex setup and management process, which includes configuring digital platforms and ensuring compatibility across systems.	Simpler process with fewer steps, primarily involving physical handling and manual tasks.

2.3.1 Importance of feedback in assessment

Table 2-1 highlighted that both assessment methods aim to provide feedback to students' submissions. C. J. Watling & Ginsburg (2019) researched the impact of feedback in assessment and found that high-quality and insightful feedback enhanced the assessment process, fostering a culture of improvement rather than a culture of performance.

Lefroy et al. (2015) summarised the key guidelines for providing effective feedback, emphasising that effective feedback must be person-oriented, timely, encouraging, specific, constructive, and actionable, to list a few. Feedback that students considered unconstructive and uncredible would often be disregarded (C. Watling et al., 2012). Any method of assessment, whether considered traditional or electronic, should aim to provide effective and constructive feedback that enriches the student's understanding, motivates them to excel in future assessments, and affirms their successes in areas where they have already demonstrated excellence.

2.3.2 Advantages of free format e-assessment over traditional assessment

It is possible to annotate and assess digital scripts without mobile technology, such as marking via a PDF editor on a computer, but it is not optimal. This alternative constrains assessors to conforming with the capabilities of the PDF editor, not allowing them to edit in free format whatsoever. Furthermore, PDF editors may not natively support question-by-question, implying that the assessor marks one question for all submissions before moving on to another question, or submission-by-submission marking, where an assessor marks each submission in its entirety, further restricting the assessor's flexibility. These limitations will prove frustrating to assessors, relying on them to place cognitive effort and thought into how they will express their feedback within the limitations of the PDF Editor.

Utilising mobile technology as a means for electronic assessment allows assessors to electronically assess and annotate in a free format, thus satisfying the criteria discussed in Section 1.4. Moreover, e-assessment via mobile technology offers several advantages over traditional assessment. It enhances efficiency by removing the need to print and store physical scripts, promoting environmental sustainability. Additionally, finding and distributing a digitally stored script is quicker than sorting through many physical scripts. Mobile devices often support stylus input, mimicking the natural handwriting and annotation process that assessors are accustomed to through traditional assessment.

2.4 Extant Systems

Section 2.2 discussed how LMSs have been implemented in higher education institutions, highlighting the implications, advantages, and disadvantages of e-assessment. This section will evaluate existing systems used for e-assessment, identifying their key strengths and weaknesses.

2.4.1 Moodle

Moodle is an open-source LMS designed for educators, students, and administrators to use within a single environment (*About Moodle*, 2024). It has a wide variety of use cases, from providing course content to students to hosting forums where students can discuss course content, assignments and ideas.

In terms of e-assessment, Moodle has various settings which can be utilised to define the parameters of any assessment as needed. Due to the nature of the problem (Section 1.2) and the required functionality (Section 1.4), the methods of e-assessment regarding manual marking with annotations of a PDF file will be explicitly evaluated, namely Moodle Gradebook and feedback via assessment files.

Moodle Gradebook

Moodle Gradebook is a component of Moodle, which includes the "Annotate Submission" feature, allowing for direct annotation through a browser. This feature enables assessors to provide direct and personalised feedback to a student's submission electronically, which students can view. This allows lecturers and students to interact with one centralised platform for assessment and feedback purposes. This feature is not limited to PDF files but can also be used for Microsoft Word (.docx), Open Document Format (.odf), and online text submissions.

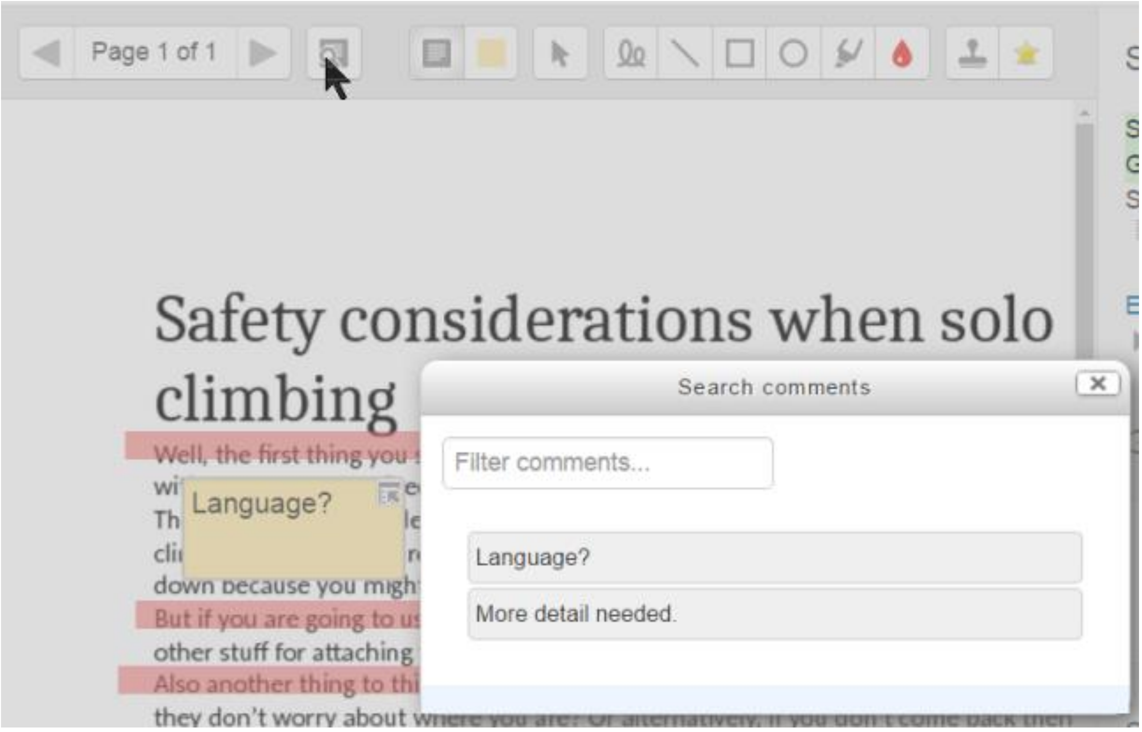

Figure 2-1: Moodle's PDF Annotation Tool

Moodle (2023) provided Figure 2-1, demonstrating an example of Gradebook's "Annotate PDF" feature. The feature provides numerous digital annotation tools, which include a pen, a highlighter tool, commenting tools, and shape drawing tools. Additionally, a "tick" feature enables users to place a tick where appropriate on the PDF.

The process of marking a submission would involve utilising the aforementioned annotation tools. After completing the assessment, the student's mark and any other necessary feedback can be recorded. Once the assessment is deemed complete, the assessor can begin marking the next submission in the queue.

One advantage of this system is its ease of use and accessibility since an assessor only requires a browser and a stable internet connection to assess submissions. However, assessors are limited to using a browser, and assessment in a constrained format is conducted solely with a mouse or keyboard. Additionally, there is no support for question-by-question marking, which can make it very tedious to evaluate submissions in a detailed manner, potentially creating an inefficient and frustrating assessment process.

Assessment via feedback files

Moodle allows submission feedback to be provided in various file formats, depending on the context. Typically, this feature is used to return a marked and annotated submission to a student. This approach offers flexibility to an assessor, allowing them to select the most appropriate format for feedback. This study will evaluate the process of providing assessment feedback via a marked and annotated PDF through Moodle.

Firstly, the assessor needs to obtain each student's submission for an assessment they wish to assess. The assessor has the option to download all submissions in a single zipped file, which they will extract to obtain the submissions for the assessment. Note that this zipped file will rename each submission so that it starts with the student's name. Figure 2-2 demonstrates an example of what an assessor may see when presented with this choice (Kahn, 2020).

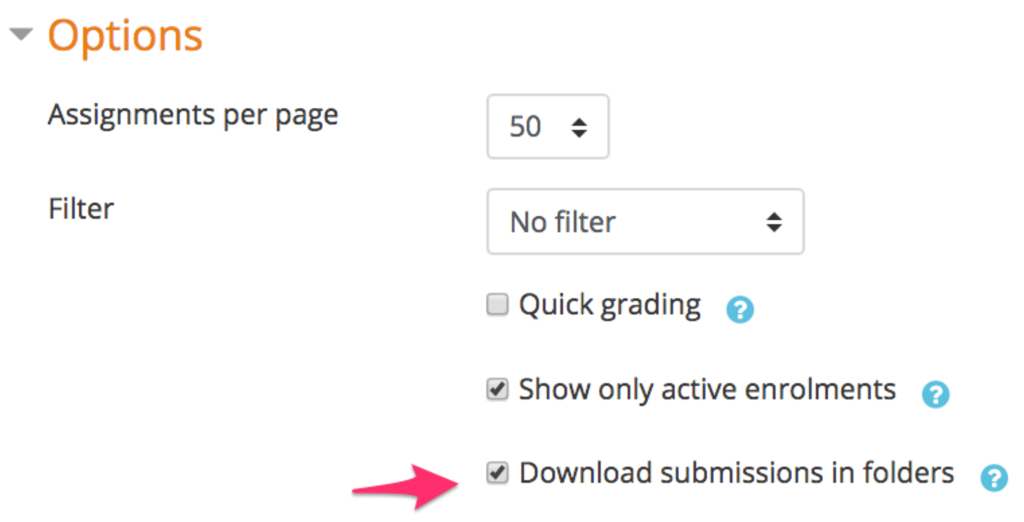

Figure 2-2: Downloading assessment submissions on Moodle

At this stage, the assessor may extract the submissions and use whatever PDF annotation tool they wish, using any device that supports such tools. After a submission is assessed, it is recommended that the assessor save the annotated script in the folder where it was accessed. If done correctly, the assessor can then zip the folder containing all the now annotated submissions and upload the zipped file back onto Moodle. Moodle will use the generated naming conventions to allocate the feedback to the correct student automatically, provided the assessor follows the correct protocol.

Although this approach gives assessors the freedom to annotate with whatever PDF annotation tool they wish. Assessors still need to record marks separately from feedback files, as feedback files contain no information other than the annotations made by the assessor. If the assessor chooses to use another device for annotations, such as a mobile device with stylus functionality, they can assess in a free format that does not constrain them. However, this creates a more tedious administrative burden, as files must be transferred between devices.

2.4.2 Blackboard Learn

Blackboard Learn is the most widely used LMS platform, offering various features for higher education institutions as of 2024 (Wheelhouse, 2024). Similar to Moodle, it has many use cases, ranging from providing course content to a tool for e-assessments (Anthology, 2022). Blackboard Learn is a commercial product and thus provides strong vendor support, which includes technical support, maintenance, and regular updates.

Blackboard Learn has several assessment opportunities, each customisable to satisfy the exact requirements any assessor might need. Figure 2-3 provides the several assessment opportunities Blackboard Learn offers, but this analysis will focus solely on Bb Annotate due to its focus on marking and annotating submissions.

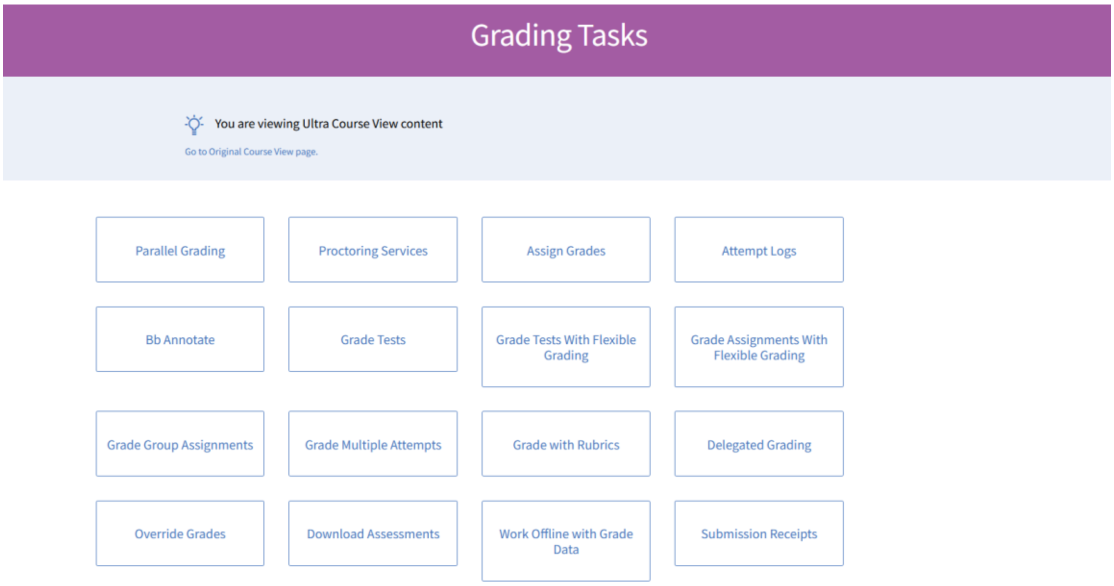

Figure 2-3: Blackboard Learn's Assessment Opportunities

Bb Annotate

Bb Annotate is a browser annotation tool that supports several document types, including PDF, Microsoft Word (.doc,.docx), Microsoft PowerPoint(.ppt,.pptx), Microsoft Excel (.xls, .xlsx), various image types, and other document formats (Blackboard Learn, n.d.). Figure 2-4 provides an example of the Bb Annotation tool that is displayed when assessing a submission.

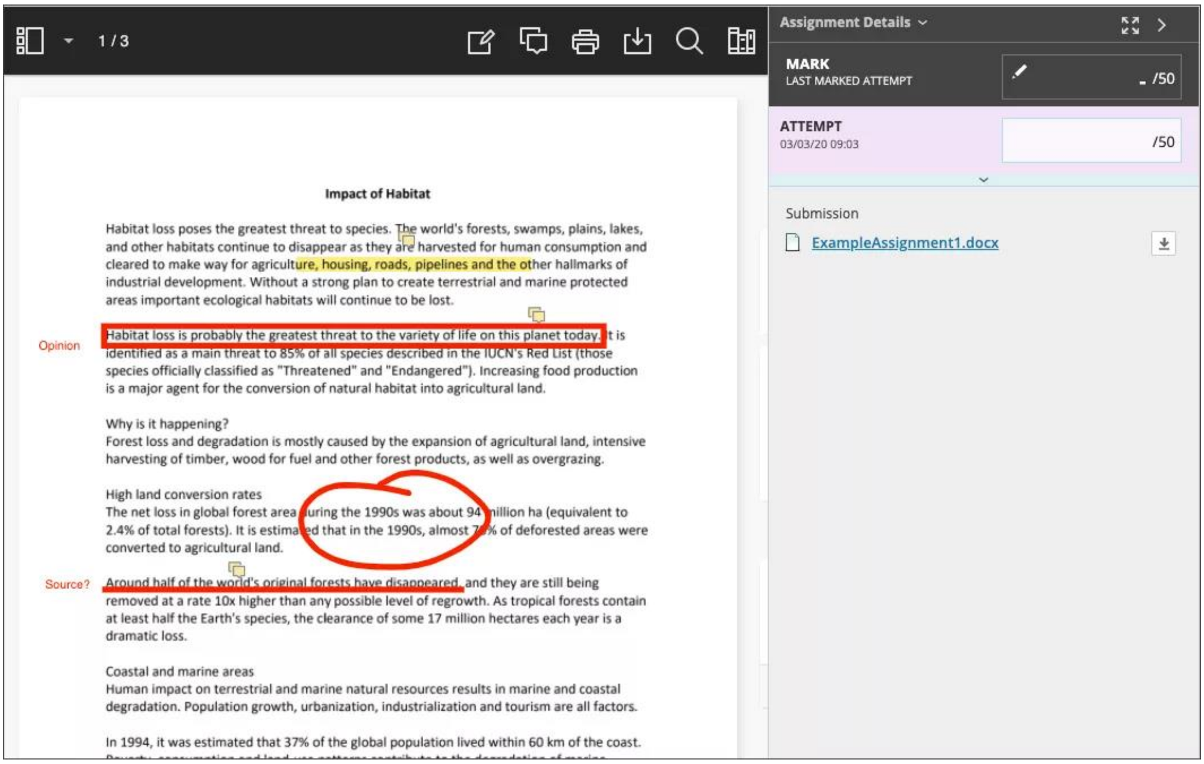

Figure 2-4: Bb Annotate Tool

Various annotation tools can be utilised, including a pen, a brush, an eraser, a highlighter, a comment box, and an undo/redo feature, among others. Reusable comments can also be added, which allows the assessor to reuse the same comment and annotation where necessary without having to retype or redraw the comment.

Assessors use these tools when grading submissions uploaded to Blackboard Learn via Bb Annotate. Similar to Moodle, the mark for the submission still needs to be recorded after the submission has been assessed. Bb Annotate has a high ease of use and accessibility because only a browser is needed to view and assess submissions, but assessment is still constrained to browser capabilities, and the lack of question-by-question support contributes to a tedious assessment process.

2.5 Conclusion

The chapter aimed to provide sufficient problem awareness, completing the first process step of the DSR methodology. This chapter discussed e-assessment, its implications and its associated advantages and disadvantages. Moodle and Blackboard Learn, two of the most widely used LMSs, were evaluated regarding their e-assessment tools for marking and annotating PDF submissions.

Evaluation of these two extant systems identified several features in the proposed solution that must also be provided and a few critical areas that current systems fall short of. The lack of question-by-question support and the additional administrative burdens diminish the potential of e-assessment by creating a more tedious and frustrating assessment process.

The shortcomings identified emphasise the need for an improved e-assessment mobile assessment and annotation tool. The proposed solution will need to incorporate the benefits of the two extant systems evaluated and the free format assessment offered by traditional assessment to create a smooth and efficient e-assessment process.

In the DSR methodology, the *Suggestion* process step follows the *Awareness of Problem* process step, involving proposing possible solutions. The knowledge gained from the analysis of extant systems will assist in the completion of the project. This new information will be vital in defining the functional requirements outlined in Chapter 3.

3 DESIGNING A MOBILE ASSESSMENT AND ANNOTATION TOOL

3.1 Introduction

Chapter 2 concluded the *Awareness of Problem* process step in the DSR methodology, as Chapter 3 progresses to the next process step, *Suggestion*, which is the main aim of this chapter. Evaluation of extant systems in Chapter 2 identified areas in which these systems excel and fall short, which will be used to identify necessary functionality for the prototype.

Beyond evaluating extant systems, requirements must be collected to identify new areas where current systems do not have any functionality or features. This approach aims to ensure that the delivered prototype has the required functionality of current systems while delivering new or improved functionality to enhance the e-assessment process.

3.2 Collecting Requirements

Schwalbe (2019) introduces the definition of benchmarking to collect requirements, which refers to comparing specific project characteristics or product features to those of other projects or products. The benchmarking was completed in Chapter 2's analysis of extant systems, which identified the following general requirements for the prototype:

- Support for annotation tools, which includes pen, eraser and highlighter functionality;
- Download and access the submitted file;
- Insert comments which can be reused for other submissions;
- Implement undo and redo functionality;
- Allocating a mark to a submission; and
- Question-by-question or submission-by-submission assessment.

These requirements have formed the base functionality the prototype will deliver. However, the prototype aims to improve the existing functionality of extant systems. Communication with stakeholders identified further areas and functionality the prototype could deliver to enhance the e-assessment process.

3.2.1 Questionnaires Issued to Assessors

Questionnaires (see *Appendix B*: *Questionnaire Issued to Lecturers*) were issued to 9 lecturers at Nelson Mandela University to identify their sentiment on the proposed system's functional requirements and any additional requirements they deem necessary. They also briefly described their marking style and were free to comment on any functionality they deemed unnecessary. The core functionality lecturers desired include:

- Automatic mark symbol recognition, which can be used to accumulate total marks per question and entire submission;
- Including a bookmarking feature, which can be utilised to assess submissions on a question-by-question basis;
- Being able to export results in a relevant format;
- Being able to publish results, either to students or moderators; and
- Automatic feedback generation, grouping common errors and trends to produce an assessment performance report.

Most lecturers agreed on the core functionality proposed to them, but an immediate area of concern was the vastly different marking techniques and approaches to assessment. However, the prototype will focus on a standardised marking technique, specifically one tick per mark and a half tick per half mark. Using the functionality identified from extant systems and the additional functionality desired by assessors, requirements have been classified into functional and non-functional requirements.

3.3 Functional Requirements

The following requirements form the basis of the prototype's expected functionality to solve the problem posed by this project. The prototype was compartmentalised into two separate components: Web-based and Mobile. This aimed to ensure that each component has a focused and distinct functionality and does not overlap unnecessarily.

3.3.1 Web-based Component

This component will be responsible for all assessment administration. Assessments can be added and edited, and final results can be published and exported. Assessed and annotated submissions can be downloaded and viewed through this component. The list below provides the core functional requirements needed by this component:

- Load a zipped file containing all submitted tasks, as PDFs, for a particular assessment into the system's database;
- Able to email selected (or all) submissions to the moderator;
- Able to email the assessment results to the moderator;
- Able to email students their individual annotated submissions; and
- Able to export the assessment results in a relevant format, such as Microsoft Excel (.xlsx) or Comma-Separated Values (.csv).

3.3.2 Mobile Component

This component will be primarily responsible for managing the core submission assessment process. Submissions for an assessment will need to be downloaded onto the device for assessment. The list below provides the core functional requirements needed by this component:

- Support for using a digital tablet to mark (with a special symbol, e.g., ✓) the tasks and add annotations to the submitted tasks in free format;
- Automatically recognise special symbols (e.g., ✓) and perform an appropriate action (e.g., in the case of ✓, accumulate a count of them and use this mark as a mark awarded for the assessment);
- Allow specific annotations/comments to be saved and reused for common errors in multiple scripts;
- Support the use of and ability to add bookmarks to choose a specific question to mark in a question-by-question manner; and
- Store the mark awarded for a particular assessment.

3.4 Non-functional Requirements

Rome (2023) defined non-functional requirements as a system's behaviour criteria. Non-functional requirements are defined by specific aspects, namely interface, performance, security, and operational. The non-functional requirements for each component will be discussed.

3.4.1 Web-based Component

Interface

The main goal of this component is to efficiently handle the administrative burden of copying submissions over to assess. Therefore, the interface needs to prioritise usability and user experience by incorporating an interface that is easy to navigate with clear and standardised navigational controls.

Performance

The system is designed to be used by one user per time, but multiple users can access this component simultaneously. Users on separate devices can utilise this component concurrently without interfering with one another.

Security

There will be various system users, such as lecturers and student assistants, each with their respective authorisation levels. For example, a lecturer can manage assessments and assess submissions, while a student assistant lacks the authority required to modify or remove assessments.

Operational

This component of the system will be developed as a web-based component, implying that device internet access and a browser will be required. Both components will interact with a server to handle concurrent users and control database interactions. This component should be compatible with any modern browser to ensure accessibility to all users. The system should be available at any time of day and function expectedly under heavy workloads.

3.4.2 Mobile Component

Interface

The main goal of this system is to increase the efficiency of annotating submissions and improve the marking experience. Therefore, the interface must promote ease of use, consistency, and memorability to ensure a memorable user experience. Additionally, the user interface should be incorporated so that the system can be utilised by novice and experienced users.

Performance

The component is designed to be used by one user per time but can have multiple users accessing the database and assessing different submissions simultaneously. It will not be possible for multiple users to assess the same submission concurrently due to conflicting marking techniques and the possibility of overwriting or removing another assessor's annotations.

Security

There will be various system users, each with their own authorisation level. To assess a submission, a user must be granted the necessary authorisation to view submissions for an assessment. By default, the module lecturer can assess submissions, but various markers can be allocated when an assessment is added using the web-based component. If a marker has been given the necessary role, they can assess a submission using this component.

Operational

This component of the system will be developed for Android tablet devices with a minimum Android API Level 29. The Mobile component will require the device's network and storage capabilities. Additionally, the Mobile component must have stylus functionality, as Section 3.3.2 outlines this as a required feature. The Mobile component will use libraries for loading and annotating PDF submissions and will be developed with Android Studio as the primary Interactive Developer Environment (IDE). This component will also work in portrait and landscape orientation and support a light and dark theme.

3.5 Initial Use Case Activity Diagram

Section 3.3 discussed the functional requirements of the prototype, compartmentalising the prototype into two components. Figure 3-1, a use case diagram, provides further insight into how actors will interact with the system's expected functionality, which is broken down into small, focused use cases.

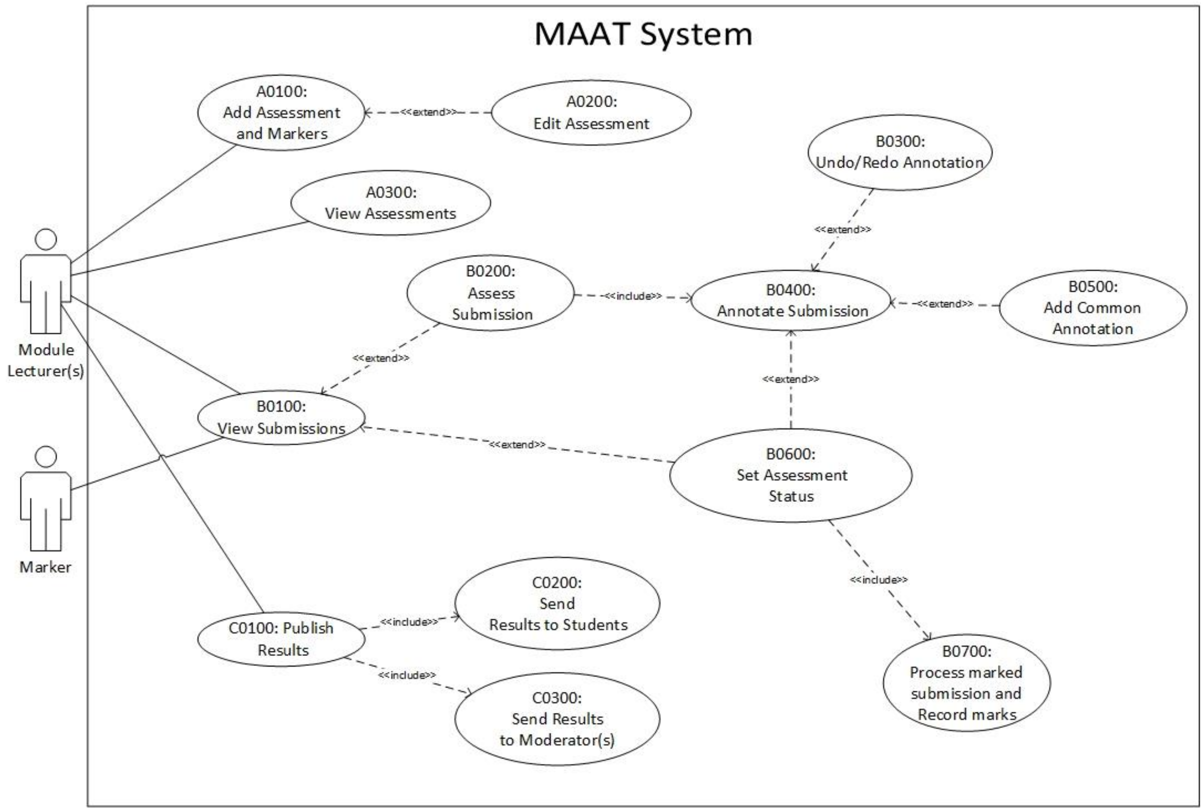

Figure 3-1: MAAT Use Case Activity Diagram

Use cases with prefixes A and C demonstrate the functionality of the web-based component, whilst the remaining use cases with prefix B demonstrate the functionality of the Mobile component. Two actors can interact with the system's functionality: module lecturer(s) and markers.

Module lecturer(s) can interact with any aspect of the system's functionality and are responsible for adding and maintaining new assessments, as well as assessing submissions. Markers can only interact with the Mobile component, meaning they can only assess submissions, provided they have the required authority level.

3.6 Initial Database Design

Following the requirements identified in Section 3.3 and the focused use cases in Section 3.5, the initial database design was created using Unified Modelling Language (UML) Database notation. There are nine tables in the database, each with their distinct purpose, which can be seen in Figure 3-2 below.

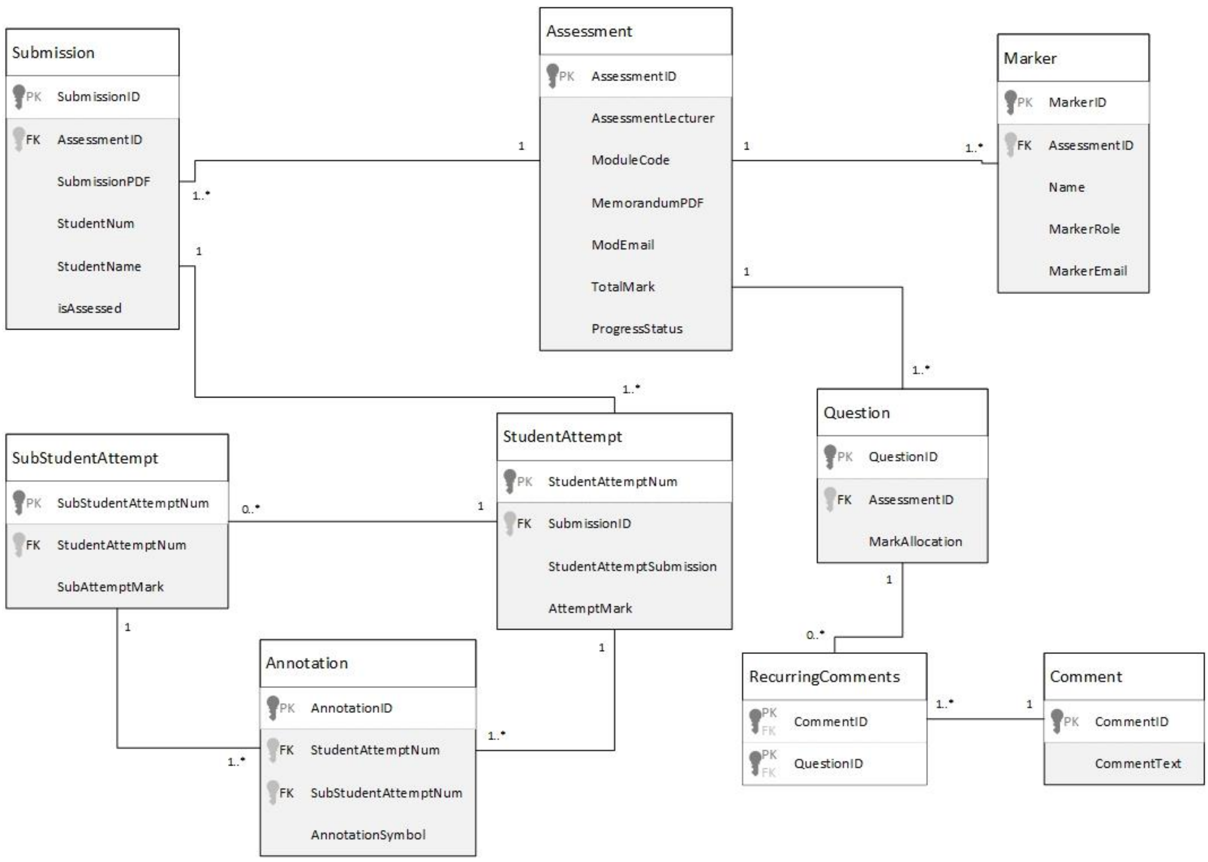

Figure 3-2: Initial Database Design

The necessary information to add to the *Assessment* table will be loaded when an assessment is added. This includes selecting the module code, uploading the memorandum, selecting the moderator, and adding the total mark of the assessment. Note that the *Assessment* table is essential in this hierarchy, as other tables cannot be created without the Assessment table being created first.

When adding an assessment, marker(s) can be selected from both Lecturers and Student Assistants to assess the submission. Module Lecturer(s) will automatically be added as markers when creating an assessment, while other Lecturers can also be assigned as markers.

Additionally, other Lecturers may be selected as Moderators for the assessment; however, a Lecturer cannot serve as both a Moderator and a marker for the same assessment. Similarly, Moderators cannot act as markers for the assessments they are moderating. Student Assistants will assume the role of Assistant and can only assess submissions using the Mobile component. The role and level of authority for each user (whether Assistant or Lecturer) is determined during the registration process, where users are explicitly registered into the system with their assigned role and corresponding permissions by the *Administrator*, who is responsible for adding, managing and removing the aforementioned users.

Adding an assessment includes uploading submissions, which will populate the *Submissions* table. As discussed in Section 2.4.1, Moodle has a naming convention when downloading all submissions, which will be analysed to obtain the student number and student name associated with a submission. Initially, isAssessed will be set to false, as the submission has not yet been assessed. When a submission is marked as assessed, isAssessed will be set to true, and the ProgressStatus in the *Assessment* table will be recalculated accordingly.

Adding an assessment includes uploading a memorandum in PDF, which will be analysed to populate the *Question* table. The assumption is that the memorandum was created in Microsoft Word, with relevant heading styles applied to certain sections, such as Heading 1 to Question 1, Heading 1 for Question 2, and so on. Provided this process was followed, the relevant information from the PDF file can be obtained to populate this table automatically. A similar approach is applied to the *SubStudentAttempt* table, which is used for sub-questions, such as Question 1.1.

Each stylus annotation is added to the *Annotation* table when assessing a submission. When an annotation is removed using the eraser or undo action, that annotation is removed from the database. This ensures that only the relevant annotations the assessor intended to remain are automatically counted per question to accumulate marks.

An assessor may save comments to be reused later, populating the *Comment* and *RecurringComments* tables. This is saved on a per-question basis, enabling assessors to apply consistent feedback to recurring errors per question in a submission. This feature aims to streamline the assessment process.

3.7 Initial User Interface Designs

Following the design of the Initial Use Case Activity Diagram, Figure 3-1 in Section 3.5 and Initial Database Design, Figure 3-2 in Section 3.6, several user interface designs were made, taking into consideration the functionality stipulated by Figure 3-1 and the data needed to be recorded by Figure 3-2. The primary user interface design per component will be discussed, beginning with the web-based component.

3.7.1 Web-based Component

Figure 3-3 below reflects the functionality of use case A0300: View Assessments. All assessments will be displayed here, those ongoing and already completed. The percentage in the centre of each circle, as well as the fill around the percentage, aims to indicate the progress of the assessment. The search bar could be utilised to filter for an assessment by name or module code.

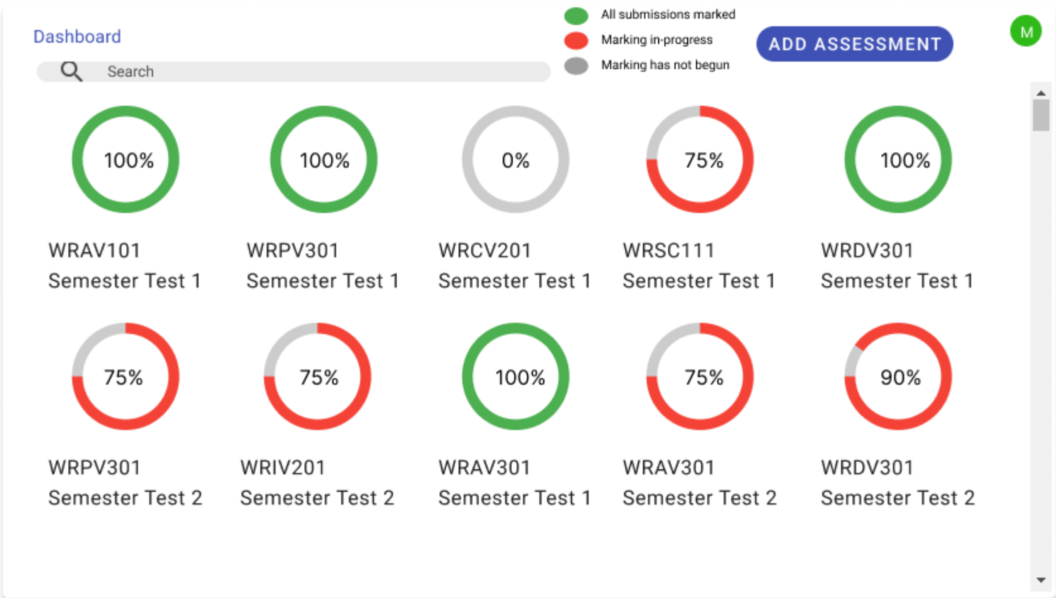

Figure 3-3: View Assessments Design

Clicking on the add assessment button will show a different design to the user, which the user can then load the necessary information to add a new assessment to the database. The progress status of an assessment will automatically update after each submission is set as marked or incomplete.

3.7.2 Mobile Component

Figure 3-4 below reflects the expected functionality of B0200: Assess Submission. This screen is where all assessing and annotating will occur. The stylus bar, positioned at the top centre of the screen, has various controls that assessors can utilise when assessing submissions. From left to right, these controls include the pen option, the eraser option, the undo option, the redo option, the text comment, and the reusable comments option.

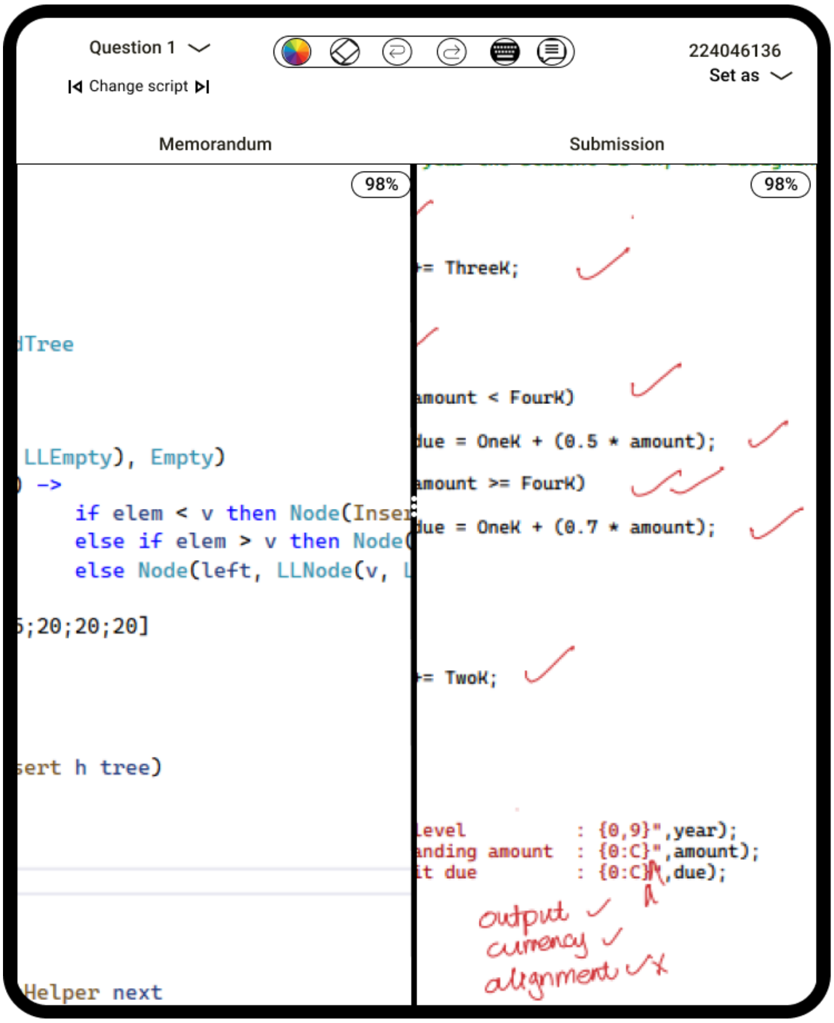

Figure 3-4: Assess Submission Design

The dropdown list in the top left contains all the questions in the submission, which selects the current question the assessor focuses on. When changing between scripts, the current selected question is saved, which supports question-by-question marking. Additionally, the split ratio between the memorandum and submission can be adjusted to the assessor's preference.

3.8 Initial Approach to Automatically Recognising Symbols

Section 3.3.2 outlined the functional requirements of the Mobile component, which identified a highly demanded requirement by assessors: automatic symbol recognition. Due to the variety and uniqueness of marking techniques, this poses a unique problem that this prototype aims to solve.

TensorFlow is an open-source platform that enables machine learning models to be built and deployed (TensorFlow, n.d.). By utilising TensorFlow, it may be possible to create a complex system using a Convolutional Neural Network (CNN) that can automatically detect handwritten marking symbols and then perform the appropriate action. The initial approach that will be used is as follows:

- Extract and segment images from the annotated PDF from the areas where handwritten annotations are placed;
- Apply preprocessing techniques to enhance the quality of the extracted images;
- Use TensorFlow to build and deploy a CNN model that contains a dataset of common marking symbols;
- Apply the CNN model to recognise and classify the handwritten marking symbols and
- Perform the appropriate action depending on the classification of the marking symbol.

This initial approach will be used to build the CNN model that automatically recognises marking symbols, thus satisfying one of the functional requirements in Section 3.3.2. This functionality will greatly enhance the effectiveness and efficiency of the e-assessment process, shifting the focus from counting marks to providing constructive and useful feedback.

3.9 Conclusion

The main aim of this chapter was to complete the second step of the DSR methodology, *Suggestion*. The prototype's functional requirements were identified through the collection of requirements completed in Section 3.2. These requirements will serve as the basis for the development of both the Mobile and web-based components.

The initial use case activity (Section 3.5), database design (Section 3.6) and user interfaces (Section 3.7) will contribute to the development of the prototype, as they were designed with the functional requirements in consideration. Additionally, the initial approach to recognising handwritten symbols (Section 3.8) will serve as a valuable guide to solving the unique problem of recognising handwritten symbols.

As the *Suggestion* process step of the DSR methodology is concluded, the DSR methodology uses an iterative approach, meaning that initial choices can be refined to meet the necessary requirements. Although the initial designs discussed in this Chapter are important, they will most likely be refined as the following chapter progresses to the net process step of the DSR methodology, *Development*.

4 DEVELOPING A MOBILE ASSESSMENT AND ANNOTATION TOOL

4.1 Introduction

Chapter 3 concluded the *Suggestion* process step of the DSR methodology, outlining the planned processes to achieve the required functionality of the proposed system. Chapter 4 progresses to the next process step, *Development*, discussing the implementation processes throughout the proposed system's development.

Chapter 3 highlighted two primary components, Mobile and Web-based, which are discussed in Sections 4.2 and 4.3, respectively. However, to facilitate communication between these components and the database, a server was introduced, which is discussed in Section 4.4. Section 4.5 outlines the implementation tools for the automatic marking symbol recognition. GitHub was used for version control in each of the aforementioned components.

4.2 Developing the Mobile Component

The Mobile Component was responsible for the assessment process of submissions, where annotations can be made through stylus input. Additional functionality was added to support offline assessment, provided the marker had at least logged in once before and had downloaded submissions to assess.

4.2.1 Implementation Tools for the Mobile Component

Table 4-1 summarises the primary implementation tools used for the Mobile Component.

Table 4-1: Implementation and Development Tools for the Mobile Component

IDE	Android Studio Koala 2024 1.1
Programming Languages	Kotlin 2.0.0 Java 17
Android API Version	Level 24 - 34
PDF Library	RadaeePDF SDK 1.2.32

Android Studio was chosen as the IDE for the Mobile Component of the proposed system due to its comprehensive suite of tools dedicated to Android app development. Additionally, Android Studio is the official IDE for Android development and provides a built-in emulator for testing and debugging. Android Studio also allows for simple deployment to physical Android devices, enabling testing outside of the built-in emulator.

Although both Java and Kotlin were used in implementation, Kotlin was selected as the primary language due to its status as the preferred language for Android app development. Furthermore, choosing Kotlin helps future-proof the Mobile component, as Android development has become increasingly Kotlin-first (Carter, 2019).

The minimum Android API was selected with a minimum SDK Version of 24 and the target SDK version of 34, the latest stable SDK version. This aims to ensure that the mobile component will be compatible with any modern hardware and to support older devices with lower SDK versions. By targeting this interval of SDK versions, the Mobile component would function with both legacy and modern devices.

The RadaeePDF SDK was used for viewing and annotating PDFs, which provided support for the complex requirements of PDF annotation through stylus input. This free and open-source version of the SDK was used in accordance with their licence and was very useful in ensuring smooth rendering and accurate annotations, which is vitally important for the proposed system. The proposed system's Android Component was primarily developed and tested on a Samsung Galaxy Tab S9 FE. Table 4-2 displays a summary of this device's specifications.

Table 4-2: Primary Development Device Specifications

Operating System	Android 13, One UI 6
Chipset	Exynos 1380 (5 nm)
CPU	Octa-Core 2.4 GHz
GPU	Mali-G68 MP5
Storage	128 GB
RAM	6 GB
Display	10.9 inches 90 Hz LCD (2304 x 1440 pixels)
Stylus (S Pen) support	Yes

4.2.2 External Libraries Used in the Android Component

Android has a built-in library management system, Gradle, which allows for simple and effective library management. The relevant dependency must simply be added, and Gradle will handle the download and installation automatically. However, custom or non-supported libraries must be added manually. *Appendix D: Android Component's External Libraries* discusses the external libraries utilised throughout the Android Component.

4.2.3 Implemented UI Designs for the Android Component

The final UI designs do not differ much in terms of functionality from the design in Chapter 3. The designs are minimalistic, aiming to be intuitive and simple to use. Figure 4-1 below shows the first screen any user sees when using the application, in which a user can log in or go to offline marking.

Figure 4-1: Mobile Log in Screen

A marker is required to log in to see the assessments they are authorised to assess—any marker logs in with their email and password with which they are registered. If a user logs in successfully, the server will return a list of assessments they have permission to assess. Alternatively, the user can choose to mark offline, allowing them to assess already downloaded submissions.

Figure 4-2 below shows the various assessments that a marker can assess. The search bar can be utilised to search for assessments by module code or assessment name. The overflow menu on the right of each assessment can be used to download all submissions for an assessment in bulk.

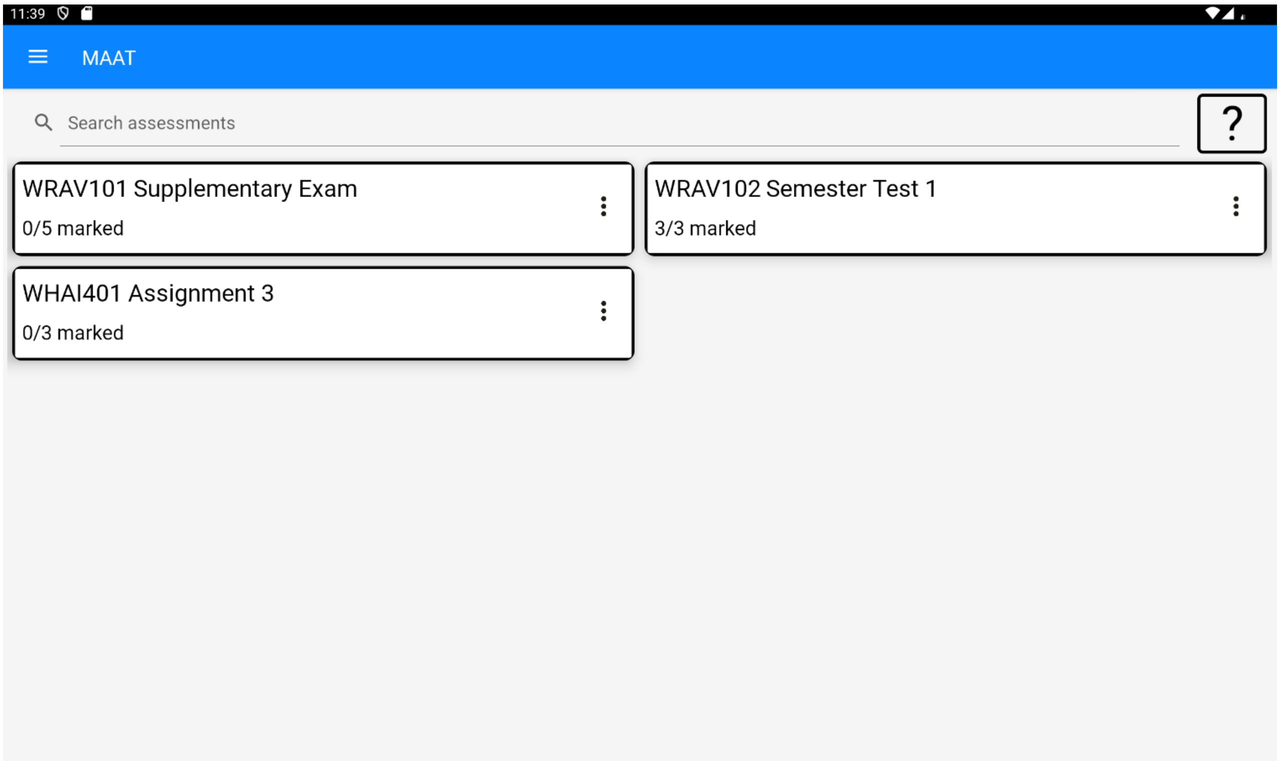

Figure 4-2: Assessments Screen

Selecting one of the assessments will load the submissions for that assessment, which is shown in Figure 4-3. Each of the submissions is set to unmarked by default, and setting a submission as marked will automatically upload that annotated submission to the server, to be saved to the database and to be marked by the Marking Symbol Recognition Component, discussed in Section 4.5.

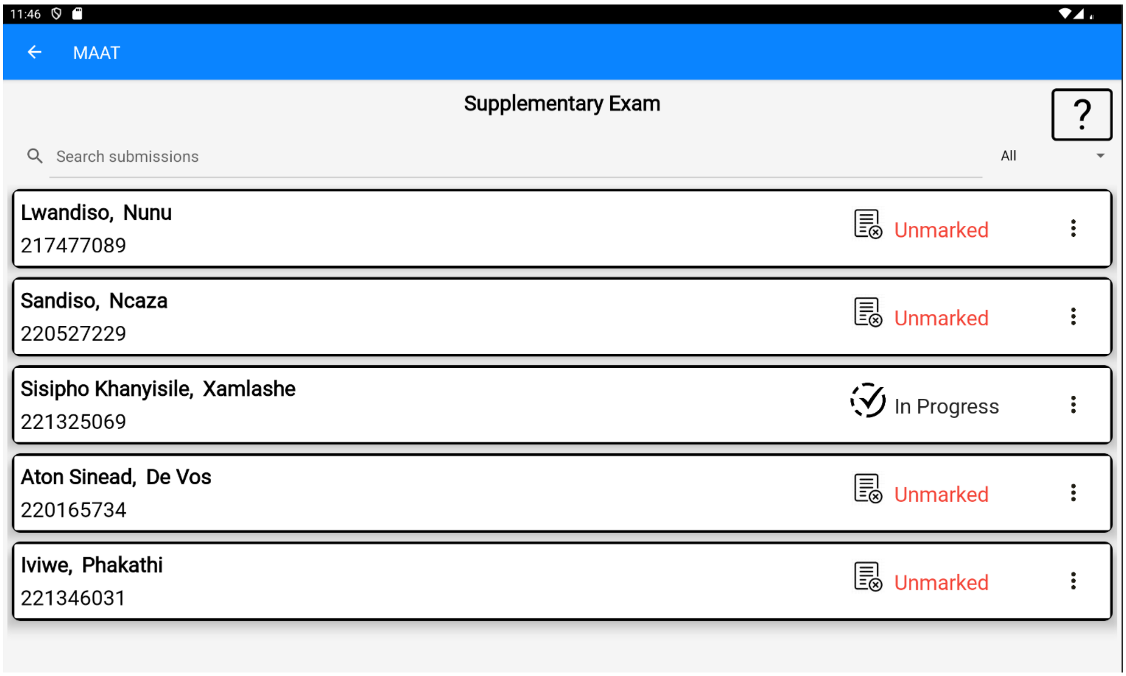

Figure 4-3: Submissions Screen

Selecting a submission to assess will first check if the memorandum and submission has been downloaded to the device. If not, then one or both documents are sent by the server and saved locally to the device. Figure 4-4 below shows the Assess Submission Screen, where the assessment process occurs.

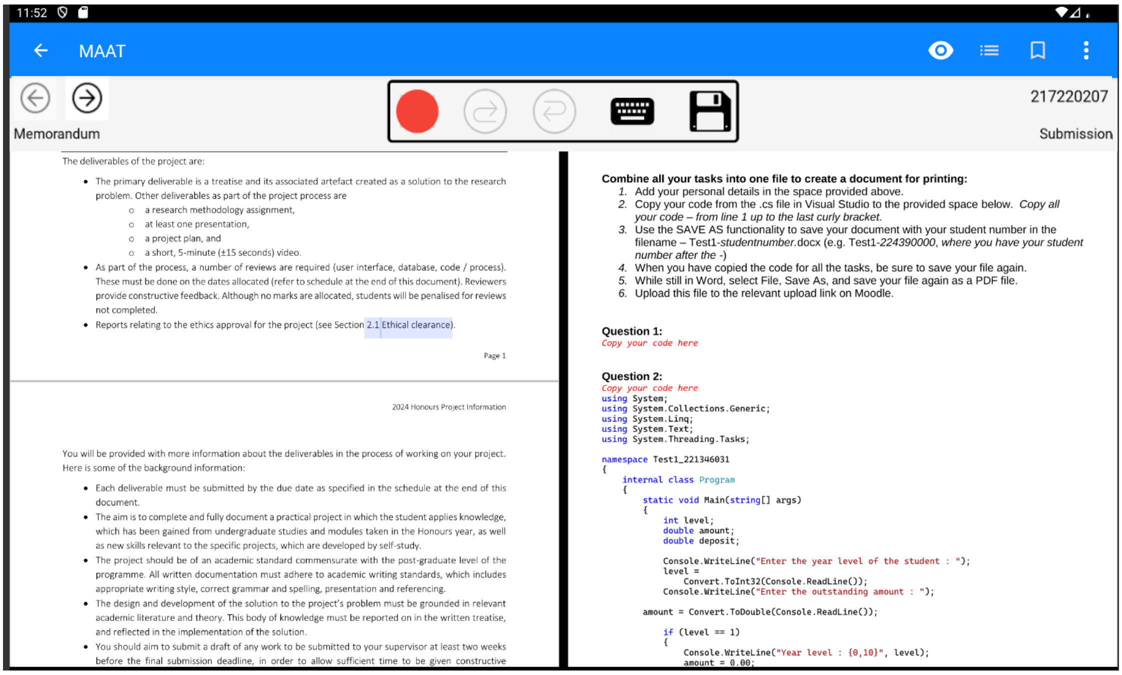

Figure 4-4: Assess Submission Screen

The Assess Submission Screen displays both the Memorandum and the Submission for that assessment, although only the submission can be edited through annotations via a stylus or typed-out comments. Various options are available in the stylus bar, which include undo, redo, add bookmark, add common annotation, add typed comment, and save edits.

4.3 Developing the Web Component

The Web Component was responsible for providing a simple and intuitive process of adding and maintaining assessments, used exclusively by Lecturers. Additionally, slight functionality was added for administrative tasks, such as maintaining modules, student assistants, lecturers, and deleting assessments.

4.3.1 Implementation Tools for the Web Component

Table 4-3 summarises the primary implementation tools used for the Web Component.

Table 4-3: Development and Implementation Tools for the Web Component

IDE	Visual Studio Code
Development Platform	Angular v18.0.6
Programming Language	TypeScript v5.4.2
Runtime environment	Node.js v20.15.0

Visual Studio Code was selected as the IDE due to its support for TypeScript and Angular development, which provided a lightweight but capable IDE for implementing the Web-based component. Additionally, Visual Studio Code also provided several extensions, such as Angular Essentials (Version 18), which simplified and enhanced the debugging and testing process significantly.

Angular v18.0.6 was selected as the Development platform due to its extensive and open-source framework for building single-page web applications that can be deployed and hosted locally for testing and demonstration purposes. This component was developed using TypeScript, which is the natively supported language for Angular Development.

Node.js was selected for the runtime environment, which allowed for simple package management through the Node Package Manager and automatically handled the backend dependencies required for the Angular framework. The Angular Command Line Interface (CLI), which requires Node.js to build any Angular application, was utilised in this component to locally deploy and host the web application.

4.3.2 External Libraries Used in the Web Component

Two libraries were utilised to obtain the required functionality or to improve the user experience. However, most required functionality was obtained through the default packages and modules provided by Angular. *Appendix E: Web Component's External Libraries* outlines the external libraries utilised in the Android Component.

4.2.3 Implemented UI Designs for the Web Component

The final UI designs do not differ much in terms of functionality from the design in Chapter 3. The designs are minimalistic, aiming to be intuitive and simple to use. Figure 4-5 below shows the first page any user sees when using the web application, in which a lecturer or an administrator can log in.

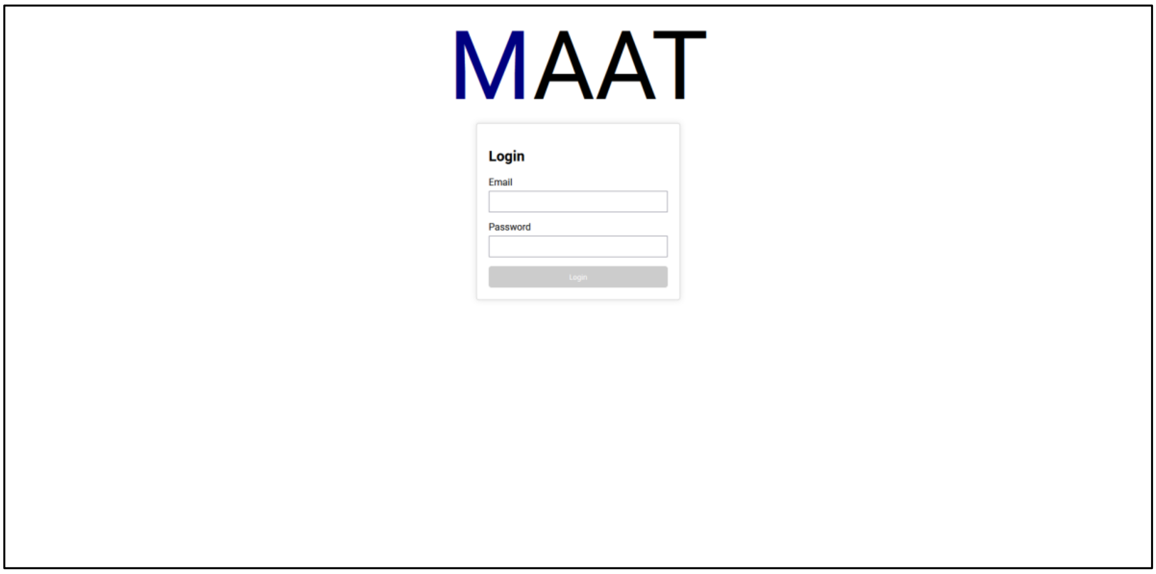

Figure 4-5: Log in Web Page

A lecturer can log in with their credentials, and if valid, they are taken to a dashboard. Alternatively, the administrator can log in, which takes them to an administrator panel, where they can manage all users and modules and remove assessments if necessary.

Figure 4-6 shows the Dashboard page where lecturers can view their current assessments. A lecturer can search for an assessment, add another assessment, or view an assessment.

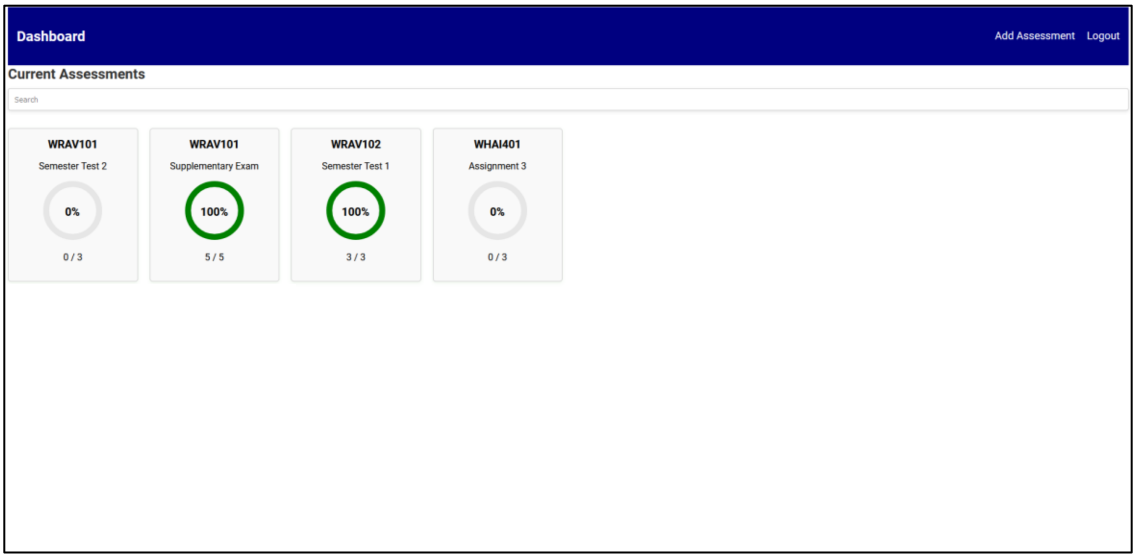

Figure 4-6: View Assessments Web Page

Clicking on an assessment will display the submissions for that assessment, some basic details, and other options, as shown below in Figure 4-7. Lecturers can publish results to students, which involves automatically sending each student their assessed script and mark. Additionally, the results are exported in CSV format, which can then be downloaded or automatically sent to the assessment moderator.

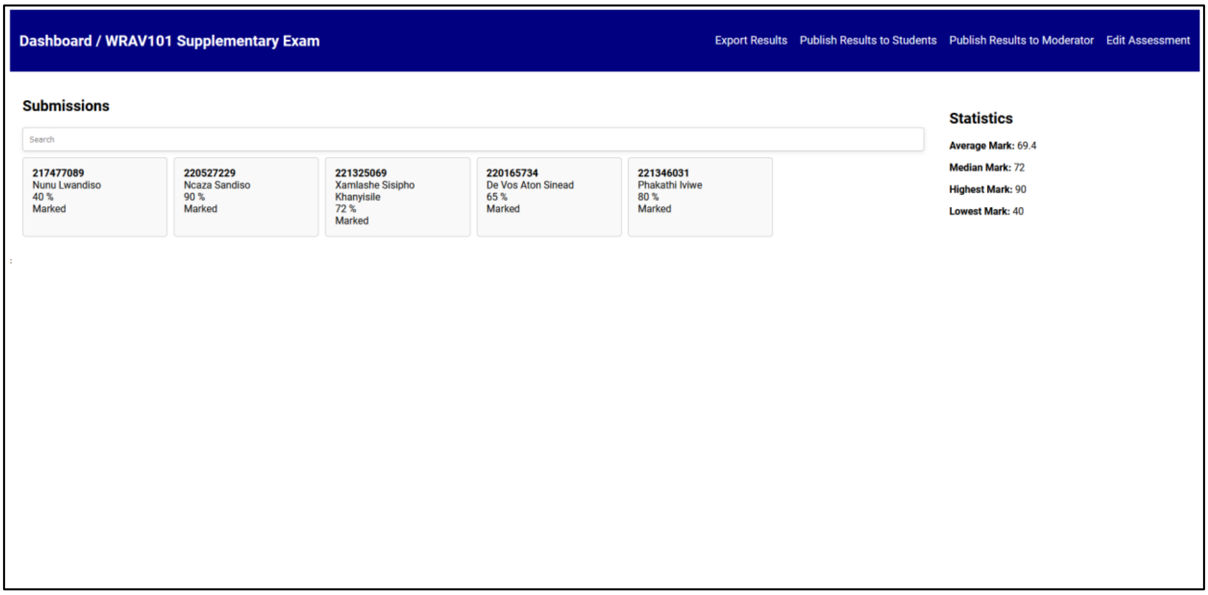

Figure 4-7: View Submissions Web Page

4.4 Developing the Server Component

To facilitate efficient communication between components and the database, a locally hosted server needed to be implemented. This server ensures that multiple users can interact with the database concurrently and allows for common logic from multiple components to be centralised.

4.4.1 Implementation Tools for the Server Component

Table 4-4 summarises the primary implementation tools used in the Server Component.

Table 4-4: Development and Implementation Tools for the Server Component

IDE	Visual Studio Code
Development Platform	Node.js v20.15.0
Programming Language	JavaScript
Runtime environment	Node.js v20.15.0

Visual Studio Code was selected as the IDE for this component due to its extensive support for JavaScript and Node.js. Consistent with the choice of IDE for the Web-based Component, Visual Studio Code is a lightweight yet powerful IDE that offers good support for the JavaScript programming language and integrates with Node.js as a Development platform.

Node.js was selected as the Development Platform (and thus the Runtime Environment) due to its ability to handle asynchronous, event-driven programming, which is ideal for developing an efficient and responsive server. Additionally, Node.js has a built-in package management system that enables additional functionality to be added by installing packages.

4.4.2 External Libraries Used in the Server Component

Node.js built-in package management system was utilised to install several packages to add the required functionality of this Server Component. Node.js has many built-in default packages, yet the additional ones that were manually installed will be discussed, as they provided the required custom functionality for the proposed system. *Appendix F: Server Component's External Libraries* discusses the various libraries utilised throughout this component.

4.5 Developing the Marking Symbol Recognition Component

This component was responsible for employing the logic required to recognise marking symbols to assess scripts automatically. This was achieved by creating a CNN trained on a dataset of ticks and half ticks to classify between these two options, done on a per-question basis.

4.5.1 Implementation Tools for the Marking Symbol Recognition Component

Table 4-5 summarises the primary implementation tools in this component.

Table 4-5: Development and Implementation Tools for the Marking Symbol Recognition Component

IDE	Visual Studio Code
Programming Language	Python v3.10.9
Runtime environment	Python v3.10.9
Machine Learning Library	TensorFlow v2.10.0

Visual Studio Code was selected for its extensive support with Python, as well as the useful extensions that enhance the testing and debugging process. Although lightweight, Visual Studio Code was a very useful IDE for various purposes among multiple components, and its versatility was very useful across multiple aspects of the proposed system.

Python v3.10.9 was utilised as the Programming Language and Runtime environment, as it was a requirement for TensorFlow v2.10.0 and had extensive support for libraries regarding machine learning and image classification. These libraries, in conjunction with TensorFlow, provided a simple and powerful method to create a neural network and train a model to classify marking symbols.

4.5.2 External Libraries Used in the Marking Symbol Recognition Component

Python has a built-in tool to install packages that were utilised to install various libraries and packages to achieve the required functionality. *Appendix G: Marking Symbol Recognition Component's External Libraries* discusses the external libraries used in this component.

4.6 Implemented Database Design

Section 3.6 outlined the initial database design, which has been altered significantly throughout development. The core database design was simplified significantly, removing redundant tables and adding various attributes that were necessary for implementation. Figure 4-8 below provides the implemented database design.

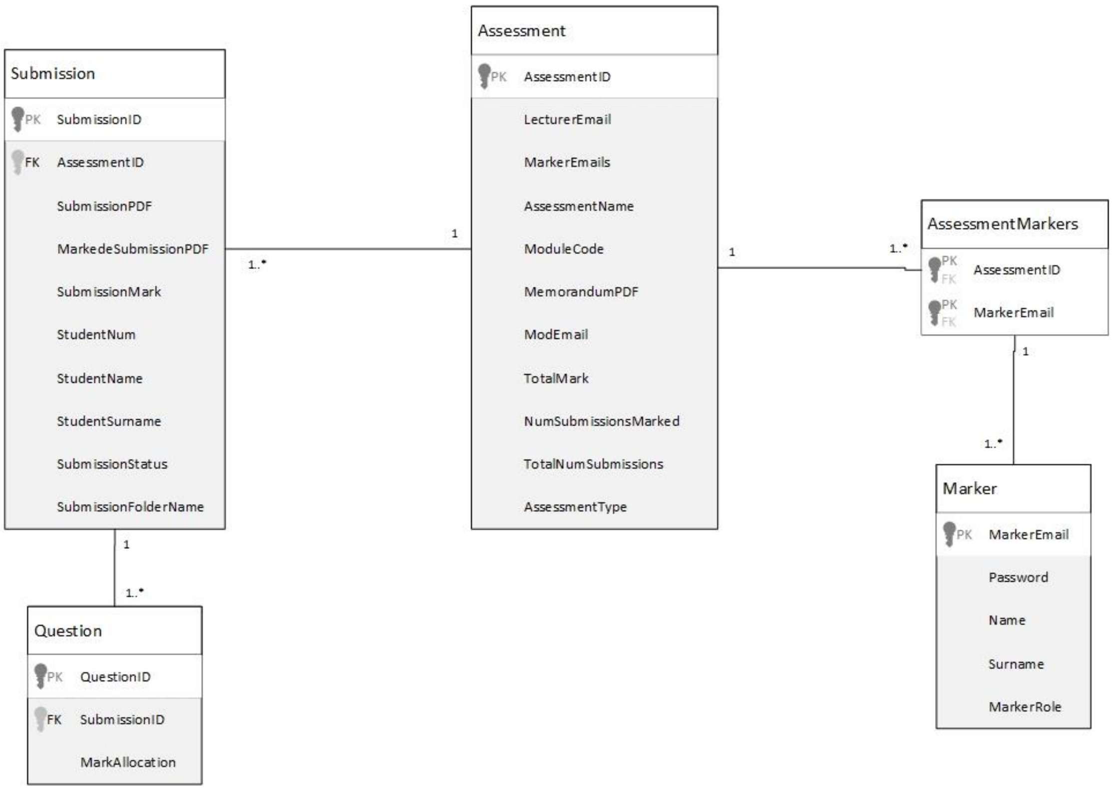

Figure 4-8: Implemented Database Design

The *StudentAttempt* and *SubStudentAttempt* tables were removed, as they provided unnecessary complexity in retrieving and maintaining student submissions. Instead, the SubmissionPDF is added directly to the *Submissions* table. The MarkedSubmissionPDF attribute is initially set to null, which is only populated when a submission has been set as marked in the Mobile Component.

Additionally, SubmissionStatus and SubmissionFolderName attributes were added to the *Submission* table. SubmissionStatus describes the progress of a submission being marked, whether it is “Marked”, “Unmarked”, or “In Progress”. Section 2.4.1 outlined how Moodle provided a consistent and uniform file structure when downloading all submissions;

SubmissionFolderName is the unique folder name for each submission, from which the name, surname and student number are automatically extracted. Additional functionality was added to cater for another type of assessment that had a different yet consistently uniform folder structure.

In the *Assessment* table, an AssessmentType attribute, such as *Moodle* or *Test Drive*, selected when adding an assessment, was introduced to differentiate between different types of assessment, each with its own unique folder structure for submissions. Each assessment type should have a consistent folder structure from which relevant information can be extracted. Additional attributes, such as NumSubmissionsMarked and TotalSubmissionsMarked, were added to display the progress of marking for an assessment.

The *Marker* table and its relationship with the *Assessment* table was altered since many markers can have many assessments. An association table, AssessmentMarker, was introduced to resolve this issue. It contained only the AssessmentID from the Assessment table and the MarkerEmail from the *Marker* table.

The *Question* table was altered to have a relationship with the *Submission* table rather than the *Assessment* table. This change was made to support question-by-question marking and is populated automatically when a submission has been set to marked. This aims to ensure that marks have been grouped per major question.

The *RecurringComments* table was removed, as the desired functionality could not be achieved utilising the RadaeePDF SDK. Section 6.2 discusses the reusable annotations feature as a recommendation for future work.

4.7 Conclusion

This chapter discussed the *Development* process of the DSR methodology, transforming the functional requirements outlined in Chapter 3 into a functional prototype. All relevant implementation tools and libraries were discussed per component to outline their functionality.

The various components, while distinct in their functionality, operate together to satisfy the requirements defined in Chapter 3. These components work in conjunction to create an integrated and cohesive system, aiming to create a more efficient assessment process for electronic submissions.

As this Chapter has concluded the *Development* process, Chapter 5 will begin the *Evaluation* process step of the DSR methodology, in which the prototype will be evaluated, and feedback will be provided to enhance further and improve the prototype.

5 EVALUATING A MOBILE ASSESSMENT AND ANNOTATION TOOL

5.1 Introduction

Chapter 4 discussed the *Development* phase of the DSR methodology, outlining the development of various components to achieve the required functionality of the proposed system, as outlined in Chapter 3. Chapter 5 addresses the *Evaluation* phase of the DSR methodology, in which the produced system was evaluated to determine whether it exhibits the required functionality, where formative evaluation and usability testing were employed for this evaluation.

5.2 Formative Evaluation

Joyce (2019) defined formative evaluation as a process utilised during the development and redesign stages to test and improve a product or interface, aiming to identify issues and areas for improvement before the prototype is completed. This formal evaluation was conducted through a code review by a senior lecturer in the Department of Computing Sciences at Nelson Mandela University during development and prior to usability testing.

The code review assessed all components of the system and found the overall functionality to be satisfactory at that time. A few recommendations were made to improve the functionality of the Mobile and Web Components:

- Add functionality to download all submissions and the memorandum for an assessment in bulk rather than individually;
- Allow offline marking so previously downloaded submissions can be assessed without needing to log in; and
- Enable exporting of all marked submissions in a zip file, preserving the folder structure of the submissions when the assessment was added to the system.

The recommendations were implemented, significantly enhancing the prototype's overall functionality.

5.3 Usability Testing

The usability testing evaluated the overall usability and user experience of the system, focusing on participant interaction with the prototype and identifying any common mistakes or confusing elements. Eight participants, all of whom complied with the Selection Criteria listed in *Appendix H: Selection Criteria*, participated in the usability testing of the prototype. Each participant was required to participate in the usability testing voluntarily and agreed to sign the consent form in *Appendix I: Consent Form*, which was required due to the ethical clearance referenced by ethics number [H24-SCI-CSS-001/U-14], in which this research project was granted, as stipulated in *Appendix C: Ethics Form*.

Each participant completed the tasks outlined in *Appendix J: Usability Test Task List*, which evaluated both the Mobile and Web Components. Once completed, the participant was required to fill in a System Usability Scale (SUS) questionnaire, which can be seen in *Appendix K: Usability Test Questionnaire*. Additionally, the questionnaire was utilised to gather more qualitative information, prompting the participants to describe anything they liked and/or disliked about the system.

5.3.1 Usability Evaluation Tasks

The participants completed the usability evaluation tasks (*Appendix J: Usability Test Task List*) independently unless help was explicitly requested. The tasks were chosen to cover key functionality of the system, ensuring that participants engaged with the most common and used features of the system.

During the evaluations, the researcher observed each participant to determine task success or failure. Any difficulties or delays encountered by participants were also observed and noted as areas for improvement. These observations, in conjunction with the SUS questionnaire, were utilised to produce results for the usability study.

5.3.2 Usability Testing Results

A comprehensive list of changes and suggestions was compiled based on the data collected from evaluation tasks and the SUS questionnaire. The changes to be implemented have been categorised per Web and Mobile Components:

- Web Component:
  - Add hover icons for interactive options, such as the “Add Assessment” Button;
  - Revise wording to clarify actions, for example, “Edit Submission” to “Save Changes”;
  - Indicate whether a lecturer is the moderator or not for an assessment;
  - Replace the dashed progress indicator of an assessment with a solid progress indicator;
  - Provide a clear visual indicator if a file has been selected during the process of adding or updating an assessment;
  - Enable partial updates for assessments, such as only editing the assessment name or total marks;
  - Introduce a distinct grid and list view when viewing the submissions for an assessment; and
  - Improve filtering, sorting and searching for assessments, markers and submissions.
- Android Component:
  - Utilise distinct undo and redo icons to clearly indicate their respective actions;
  - Indicate to the user that the divider in the centre of the Assess Submission screen that the submission and memorandum views are resizable;
  - Utilise the same help functionality in the action bar (bar at the top of each screen) of each activity;
  - Allow for the marker to see the submission mark if the status is marked;
  - Replace the red circle icon with a pen icon to indicate stylus input; and
  - Clearly indicate the status of a button and whether or not it is interactive.

An additional change was recommended for future consideration (Section 6.2). The suggestion involves filtering out available markers and lecturers based on the selected module when adding or editing an assessment in the Web Component.

The SUS questionnaire (*Appendix K: Usability Test Questionnaire)* was utilised to calculate an SUS score for each participant (Albert & Tullis, 2023). For questions 1, 3, 5, 7 and 9, the SUS score contribution is derived by subtracting one from the participant's response. For questions 2, 4, 6, 8 and 10, the contribution is derived by subtracting the participant's response from 5. The sum of these values is then multiplied by 2.5 to provide the SUS score per participant. Table 5-1 highlights the SUS score per participant, as well as the average score.

Table 5-1: SUS Questionnaire Results

Participant	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Question 1	4	5	4	5	5	5	5	5
Question 2	1	2	2	1	1	1	1	2
Question 3	5	5	4	5	4	5	5	5
Question 4	2	2	2	1	4	1	1	1
Question 5	5	5	5	5	5	5	5	5
Question 6	1	1	1	1	1	1	1	1
Question 7	5	5	5	4	4	5	5	4
Question 8	1	1	1	1	2	1	1	1
Question 9	4	5	5	5	4	5	5	5
Question 10	1	1	1	1	3	1	1	1
SUS Score	92.5	95	90	97.5	77.5	100	100	95
Average SUS Score	93.44							

Albert & Tullis (2023) stated that an average SUS score of greater than 70 is considered acceptable. Further analysis by Bangor et al. (2009) offered a grading scale interpretation, where a SUS score between 90 and 100 was graded as an "A", the highest obtainable grade. An average SUS score of 93.44 is very promising for an initial evaluation, and implementing suggested improvements could further enhance the system's usability score.

5.4 Implementation Changes for the Web Component

Section 5.3.2 outlined several minor changes to be implemented in terms of user interface and functionality to further enhance and streamline the user experience of the Web Component. Figures 5-1, 5-2 and 5-3 demonstrate most of these minor UI and functionality changes.

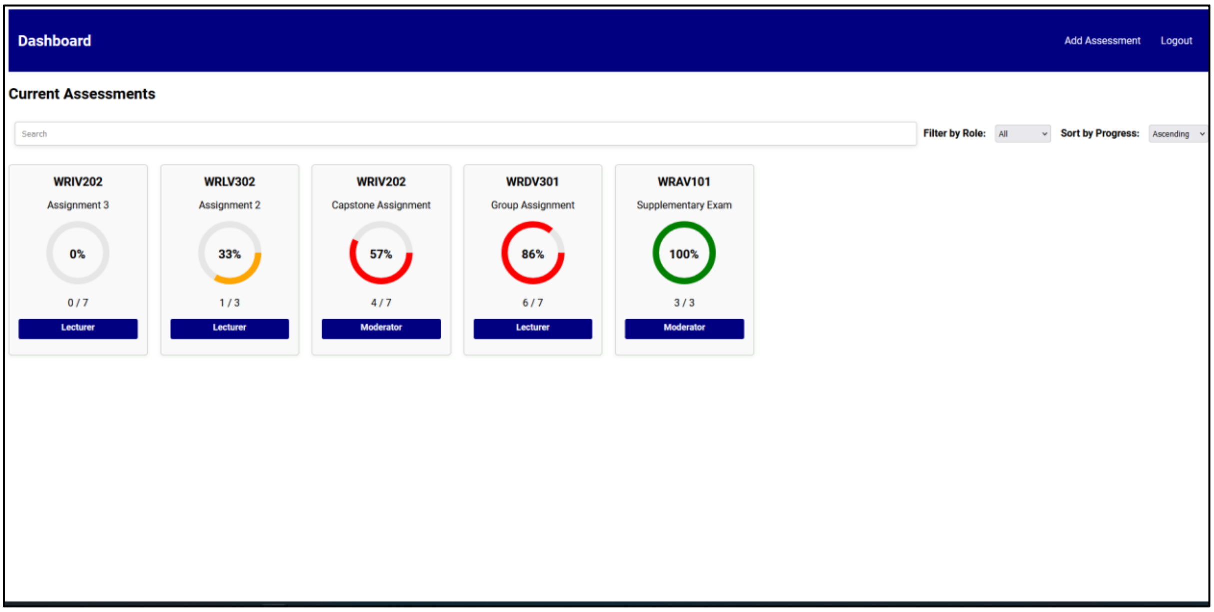

Figure 5-1: Updated View Assessments Web Page

Figure 5-1 above shows the updated View Assessments Web Page, with a label applied to each assessment to clearly indicate a user's role for an assessment, either a *Lecturer* or *Moderator*. Moderators have the same permissions and functionality as Lecturers when viewing an assessment, meaning they can view and alter submission results, edit an assessment, and perform other actions. This alteration aimed to ensure clarity for Lecturers and Moderators among their respective roles for various assessments.

Additional functionality was added to allow users to filter for the assessments for which they are either the Lecturer or Moderator and sort in either descending or ascending order in terms of marking progress, which is indicated by a percentage for each assessment. Furthermore, the dashed progress indicator in the centre of each assessment was replaced with a solid progress indicator. The solid progress indicator provides a more natural and intuitive representation of the marking progress of an assessment and better represents the percentage indicator in the centre of each circle.

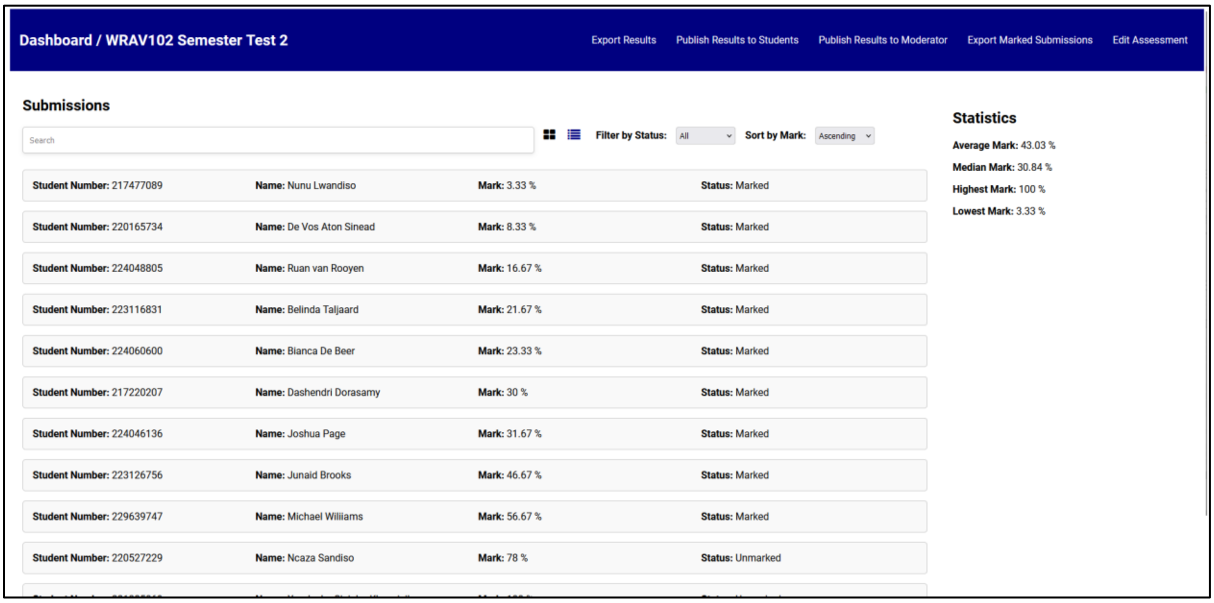

Figure 5-2: Updated View Submissions Web Page

Figure 5-2 above demonstrates the updated View Submissions Web Page, with the list view applied instead of the default grid view. Additional functionality was added to filter submissions based on status and sort submissions on the mark.

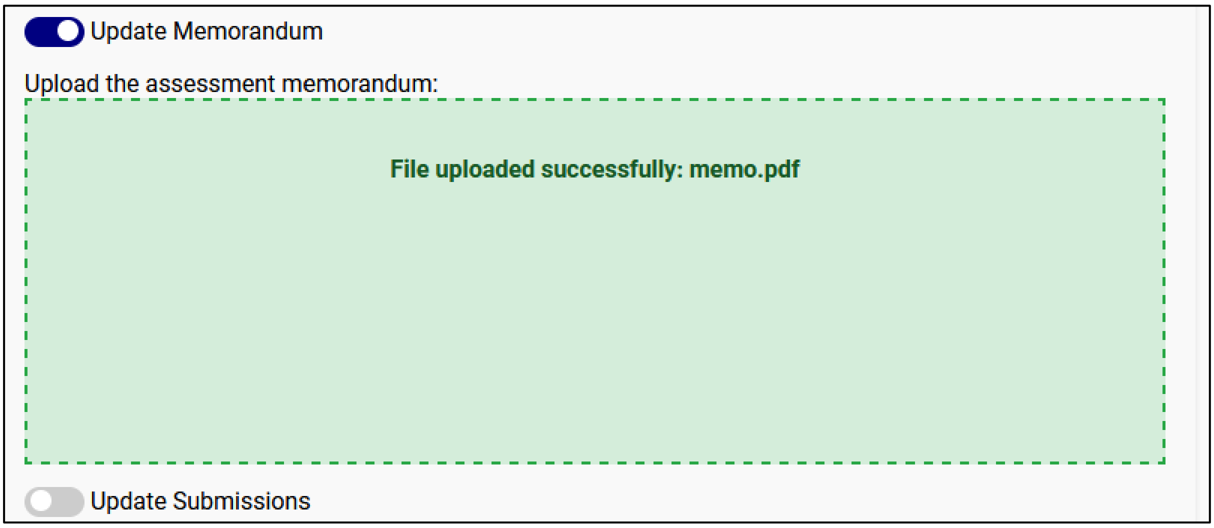

Figure 5-3: Updated UI elements for uploading files

Figure 5-3 above illustrates the updated interface element for uploading memoranda and submissions, used when updating or adding an assessment. The interface now clearly indicates when a file has been successfully uploaded. Additionally, the toggles for "Update Memorandum" and "Update Submissions" demonstrate the added functionality for partial editing of assessments.

5.4 Implementation Changes for the Mobile Component

Section 5.3.2 outlined a few changes in terms of user interface and functionality for the Mobile component, primarily for the Assess Submissions Screen. One element that has been altered across all screens is the alteration of the help button's position. Each screen offers a help option, in which the purpose and intended outcome of each screen are explained briefly if the user chooses to interact with said help option. All of these changes are demonstrated in Figures 5-4 and 5-5.

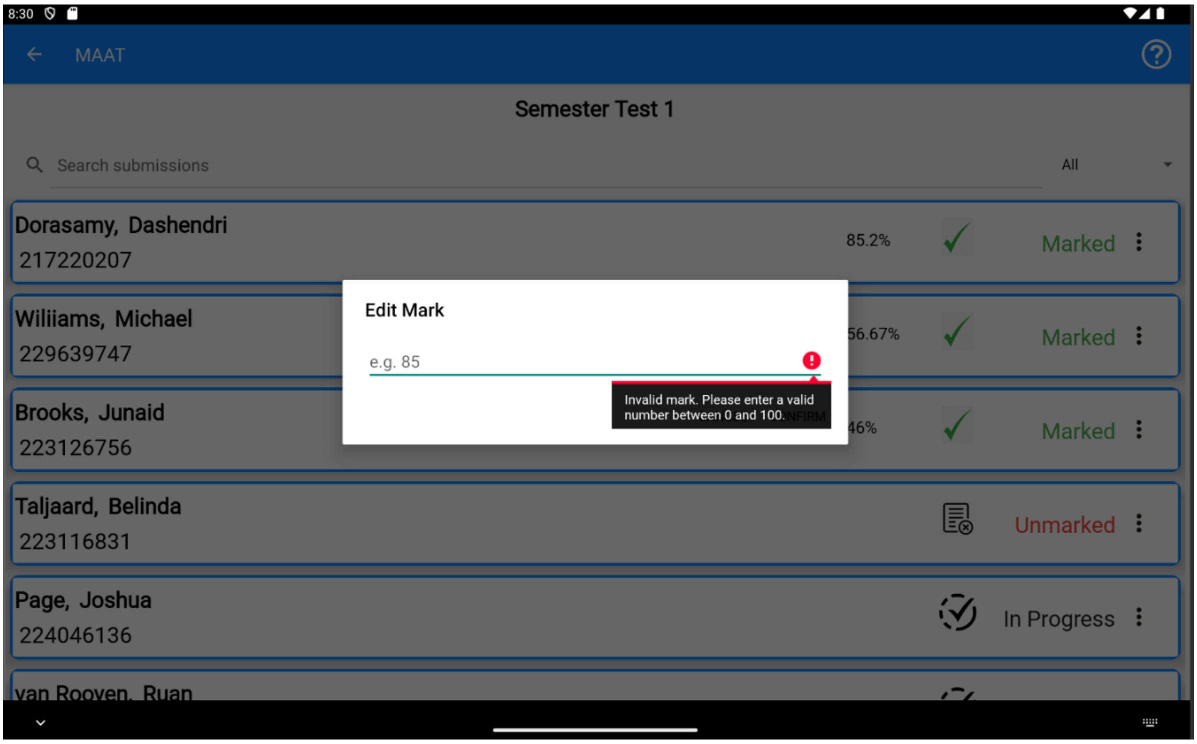

Figure 5-4: Updated Submissions Screen

Figure 5-4 illustrates the updated Submissions Screen, with the only addition being the submission mark that appears if a submission has been marked. When the marker clicks on the submission mark, an alert dialogue box appears, enabling them to adjust the submission mark, if necessary, which is also demonstrated above. To ensure data validation, only a valid mark between 0 and 100 is accepted; otherwise, a brief error message will be displayed.

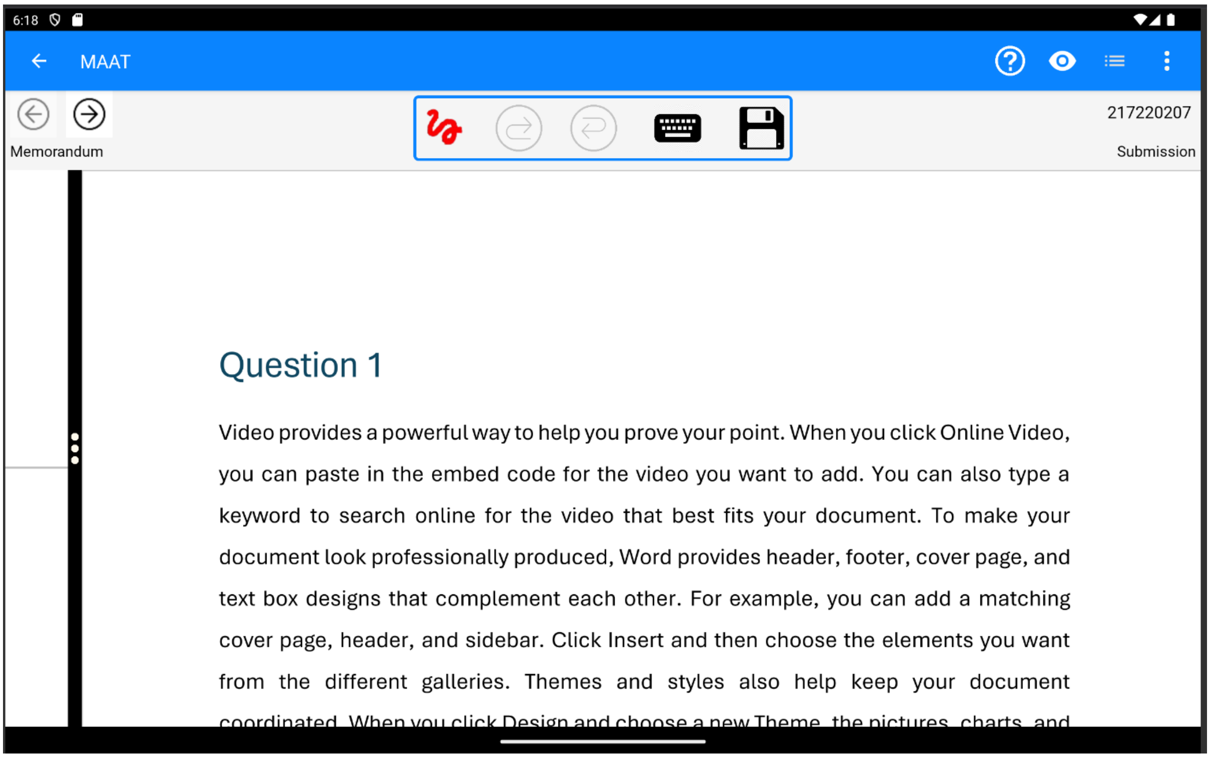

Figure 5-5: Updated Assess Submission Screen

Figure 5-5 illustrates the updated Assess Submission Screen based on the various suggestions. If a button cannot be interacted with at the current moment, it is greyed out to indicate its inactive state, as illustrated by the undo and redo buttons. Additionally, the red circle icon was replaced with a more intuitive icon to better indicate the stylus functionality.

To assist users who are unsure about the actions of each button, a help button has been added, indicated by the question mark. When clicked, an alert dialogue box appears, explaining the functionality of each icon. Furthermore, if a user long-clicks any button in the stylus bar, an alert dialogue box will provide a detailed explanation of that button's function.

5.5 Conclusion

This chapter discussed the *Evaluation* process of the DSR methodology, identifying various suggestions and areas of improvement through formative evaluation and usability testing, which focused on both the functionality and user experience to ensure a comprehensive assessment of the prototype. Although various areas of improvement were identified, the average SUS score indicated a positive overall user experience, with some opportunities for enhancement that could further improve and enhance the overall user experience.

Since the post-usability test implementation of changes for both the Web and Mobile Components have been made, it is reasonable to conclude that the development of the prototype has been completed. Consequently, the project aims (Section 1.3) have been met, and the system exhibits the required functionality and enhanced user experience.

As this Chapter has concluded the *Evaluation* process, Chapter 6 will begin the *Conclusion* process step of the DSR methodology, in which future considerations and ideas will be discussed that may further enhance and improve the system.

6 CONCLUSION

6.1 Introduction

The primary aim of this project was to develop a Mobile Assessment and Annotation Tool to enhance and streamline electronic assessment whilst combining the advantages of both electronic and traditional assessment. DSR Methodology (Section 1.5) was utilised throughout the project lifecycle and provided a valuable stsructure and iterative approach for research and development. Chapter 2 researched the domain of electronic assessment and evaluated two extant systems, identifying their weaknesses and shortcomings. Chapter 3, in conjunction with Chapter 2, established the basis in terms of functional and non-functional requirements, as well as outlining the process of how to achieve the aforementioned functional and non-functional requirements.

Chapter 4 discussed the implementation of each component, highlighting the approach taken to achieve the required functionality as stipulated in Chapter 3. Chapter 5 evaluated the prototype through formative evaluation and usability testing, and although the initial SUS score was good, several minor suggestions for improvement were highlighted and subsequently implemented.

6.2 Recommendations for Future Work

Although the system has satisfied the project aims (Section 1.3), there are still areas and aspects the system can be expanded on to provide new functionality or improve upon the existing functionality of the prototype. The following suggestions are intended to enhance the prototype by expanding on functionality and making it applicable to a broader range of assessors:

- Change the IDE for the Mobile version to Flutter (Shivanandhan, 2023) or React Native (Sachan, 2024), which will allow for deployment on Android and iOS operating systems;

- Add reusable annotations, which was a functional requirement outlined in Chapter 3 (Section 3.2) but was unable to be implemented due to the constraints of the RadaeePDF SDK library;
- Integrating with an LMS, such as Moodle and Blackboard (discussed in Section 2.4), to allow for submissions and assessments to be added automatically;
- Expand upon the marking symbol recognition, such as being able to recognise a tick with a number next to it, indicating that said tick is worth the number of marks;
- Implementing plagiarism detection, either using a custom technique or utilising the Turnitin API (Chu, 2021); and
- Produce an automated feedback file for assessors, which may highlight weak areas or common mistakes for many submissions across an assessment.

While the prototype can still be improved, this project has achieved the desired functionality as stipulated in the project proposal (see *Appendix A: Project Proposal*) and the non-functional and functional requirements (Sections 3.3 and 3.4), with the exception of the removed reusable annotations feature.

6.3 Reflections

Initially, this project seemed a daunting experience to me; much was expected of me, and time seemed to move so quickly, particularly in development. I had very limited experience with using Kotlin as the primary programming language for the Android Development environment, as well as little to no experience utilising TypeScript, JavaScript and Python programming languages for the various components. Implementing the Web Component using Angular and Typescript was a challenge for me, as I had no experience creating interactive and dynamic websites before.

6.3.1 Insights

This project provided valuable insights into my approach to multi-tasking, as well as the process of researching. Initially, I was hesitant to engage in research, but I found it a valuable and insightful experience; without it, I would have produced a worse system for it. Working with multiple development environments, as well as completely different components, and utilising the server to integrate them all together was a challenging yet deeply rewarding experience.

Furthermore, this project has taught me to become a much better programmer, especially in optimisation and abstraction, allowing me to utilise object-oriented programming effectively to produce a functional and easily extendable system. Additionally, working with another library, namely the RadaeePDF SDK library, assisted me in the challenging process of reading and understanding code that is not mine; patience and consistency were the key factors in this regard.

Lastly, my time and project management skills have greatly improved. Learning how to manage my time effectively, as well as plan what I need to achieve before attempting to achieve it, was a helpful and valuable experience; I have learned it is often more effective to thoroughly think through an approach before proceeding with no planning at all. I am very proud of the research and system I have produced under the guidance of my supervisor, and I am thankful for this research project experience.

6.3.2 Challenges

This project had many obstacles throughout development, particularly the constraints of the open-source RadaeePDF SDK library that was used for viewing and annotating PDFs. This library was very complex and had poor documentation, so much time was needed to understand exactly how the library functioned and how to utilise it for the established functional and non-functional requirements.

The time taken to learn how to utilise new programming languages, development environments, concepts, tools, and research skills effectively was another challenge in itself; time always seemed against me. However frustrating, they provided the necessary functionality to achieve the project aims.

The most challenging feature was the attempt to implement reusable annotations, which was a functional requirement but was removed as I could not achieve this requirement. Working with this library was very challenging, and this was by far the most frustrating aspect of development; it has now become a suggestion for future work (Section 6.2).

6.4 Conclusion

The primary aim of the project was to develop a functioning tool for Mobile Assessment and Annotation whilst providing a similar experience to traditional marking techniques and combining the advantages of both electronic and traditional assessment (Section 1.3). This was done by researching various extant systems and discussing their shortcomings, as shown in Chapter 2. The required functionality of the system was finally identified using requirements collected through questionnaires issued to lecturers, as well as the identified shortcomings of investigated extant systems. These findings formed the basis of functional and non-functional requirements in Chapter 3 and planned the process of implementation outlined in Chapter 4. Chapter 5 discussed the evaluation of the prototype, achieving a high SUS score, indicating that the aims of this project have been met.

In conclusion, it can be said that the research obtained throughout the development of this project has been added to the research base in the domain of assessment. This prototype has confirmed that a Mobile Assessment and Annotation Tool can be utilised to achieve the benefits of both electronic and traditional assessment, removing the frustrating and tedious aspects related to assessment.

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A P P E N D I C E S

Appendix A: Project Proposal

2024/16

Title:	**Mobile Assessment and Annotation Tool**
Proposer:	M Taljaard
Potential Supervisor(s):	Names of supervisors who would be suited to supervise the project.
User(s):	M Taljaard, D Vogts, MC du Plessis
Pre-requisite(s):	Modules required to take the project.
Best suited for degree	☒BSc ☒ BCom ☐ BCom IS ☐ Not specific
Implementation tool(s):	To be determined by student in collaboration with supervisor(s)
Student:	Has this project been allocated to a student? If so, who?
Priority:	What is the priority of the project?
Comments:	Possibility for this project to feed into a more advanced project related to the analysis of data logged from the annotation process.
Complexity:	Slightly complex
Keywords:	Assessing programming, coding, coding feedback, annotating documents
Problem Statement:	Development of a system that makes use of a digital tablet to overlay marking annotations onto online submisisons and manages special notations made by the assessor while electronically assessing the online submissions.

Description:

Programming modules like WRAV1, WRAV2 and WRPV3 rely on the assessment of online solutions during the process of summative assessments (practical tests). There are two ways in which these submissions are manually marked: a) by inspection of the tasks on the screen and the assignment of a mark, or b) by printing out the submission and marking the hard copy version. Each of these methods has its benefits and shortcomings.

Relying only on the digital version does not require any additional management of printing resources. Despite this advantage, this technique of assessment does not encourage the use of formative feedback at appropriate places in the submitted tasks and therefore the student cannot benefit in the best possible way. Marking on a hard copy version requires the additional management of printing resources during assessment times, but the assessor has the opportunity to make annotations on the hard copy at appropriate places for viewing by both moderators and specifically students.

Mobile technology allows for making use of a stylus to indicate marks and comments on a pdf version of a submission. But ideally a tool should include options that can improve the marking experience for the lecturer, and to enable the lecturer to provide constructive feedback to the student in an easy way.

There is thus a need for a system that can successfully combine the benefits of the two techniques with additional features to make the assessment of online submissions more usable and the process surrounding these assessments more productive.

The system would have at least the following functionality:

- Load all submitted tasks for a particular assessment into the system's database;
- Support for the use of a digital tablet to mark (with a special symbol e.g. ✓) the tasks and add annotations to the submitted tasks in free format;
- Save the annotated tasks for later reference by the moderator(s) and relevant student;
- Automatically recognize special symbols (e.g. ✓) and perform an appropriate action (e.g. in the case of ✓, accumulate a count of them and use this as a mark awarded for the assessment);
- Store the mark awarded for a particular assessment and support the exporting of this data in a format that is compatible with an Excel spreadsheet;
- On request from the assessor, email selected (or all) assessments to the moderator; and
- On request from the assessor, email to each student his/her annotated assessment.

Honours Project Proposals

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Appendix B: Questionnaire issued to Lecturers

Mobile Assessment and Annotation Tool

Thank you for agreeing to participate in this questionnaire. The primary goal of this questionnaire is to identify any additional requirements that lecturers may want from a Mobile Assessment and Annotation Tool, as well as identifying current marking styles the tool needs to cater for.

* Indicates required question

1. **Please enter your name and surname**

____

2. **Please discuss your marking style (one mark per tick, tick can be equivalent to multiple marks, etc.)** *

____

3. **Which of the following features would you use?** *

Check all that apply.

- [ ] Automatic symbol recognition - Recognising special symbols and then performing the appropriate action, while accumulating a count of marks awarded.
- [ ] Reusable annotations/comments - Ability to save and reuse specific annotations/comments for reoccuring errors across multiple scripts.
- [ ] Bookmarking - Support for adding bookmarks to specific questions or areas of interest in a single script
- [ ] Publishing results - Email selected or all scripts to students and/or the moderator
- [ ] Exporting results - Exporting results to relevant formats (e.g., pdf, .xlsx.)
- [ ] None

4. **Which of the following features would you not use?** *

Check all that apply.

- [ ] Automatic symbol recognition - Recognising special symbols and then performing the appropriate action, while accumulating a count of marks awarded.
- [ ] Reusable annotations/comments - Ability to save and reuse specific annotations/comments for reoccuring errors across multiple scripts.
- [ ] Bookmarking - Support for adding bookmarks to specific questions or areas of interest in a single script
- [ ] Publishing results - Email selected or all scripts to students or the moderator, or both
- [ ] Exporting results - Exporting results to relevant formats (e.g., pdf, .xlsx.)
- [ ] None (I would use all)

5. **What additional features would you like to see implemented?** *

Appendix C: Ethics Form

NELSON MANDELA
UNIVERSITY

PO Box 77000, Nelson Mandela University, Port Elizabeth, 6031, South Africa mandela.ac.za

Chairperson: Research Ethics Committee (Human)
Tel: +27 (0)41 504 3624
Dalray.Gradidge@mandela.ac.za
NHREC registration nr: REC-042508-025

Ref: [H24-SCI-CSS-001/U-14] / Approval: 28 March 2024 – 28 March 2025

28 March 2024

Ms M Taljaard
Faculty: Science

MOBILE ASSESSMENT AND ANNOTATION TOOL

PRP: Ms M Taljaard
PI: Mr J Page

Your above-entitled application served at the Research Ethics Committee (Human) (meeting of 27 March 2024*)* for approval. The study is classified as a medium risk study. The ethics clearance reference number is **H24-SCI-CSS-001/U-14** and approval is subject to the following conditions:

1. The immediate completion and return of the attached acknowledgement to Imtiaz.Khan@mandela.ac.za,
2. Approval for data collection is for 1 calendar year from date of receipt of this ethics approval letter.
3. The submission of an annual progress report by the PRP on the data collection activities of the study (form RECH-004 available on Research Ethics Committee (Human) portal) by 15 November this year for studies approved/extended in the period October of the previous year up to and including September of this year, or 15 November next year for studies approved/extended after September this year.
4. In the event of a requirement to extend the period of data collection (i.e., for a period in excess of 1 calendar year from date of approval), completion of an extension request is required (form RECH-005 available on Research Ethics Committee (Human) portal)
5. In the event of any changes made to the study (excluding extension of the study), RECH will have to approve such amendments and completion of an amendments form is required PRIOR to implementation (form RECH-006 available on Research Ethics Committee (Human) portal).
6. Immediate submission (and possible discontinuation of the study in the case of serious events) of a report to RECH in the event of any unanticipated problems, serious incidents or adverse events observed during the course of the study.
7. Immediate submission of a Study Termination Report to RECH (form RECH-008 available on Research Ethics Committee (Human) portal) upon expected or unexpected closure/termination of study.
8. Immediate submission of a report to RECH in the event of any study deviations, violations and/or exceptions.
9. Acknowledgement that the study could be subjected to passive and/or active monitoring without prior notice at the discretion of Research Ethics Committee (Human).

Please quote the ethics clearance reference number in all correspondence and enquiries related to the study. For speedy processing of email queries (to be directed to Imtiaz.Khan@mandela.ac.za), it is recommended that the ethics clearance reference number together with an indication of the query appear in the subject line of the email.

We wish you well with the study.

Yours sincerely

DGradidge

Dr D Gradidge
Chairperson: Research Ethics Committee (Human)

Cc: Department of Research Development
Faculty Admin: Science

Appendix 1: Acknowledgement of conditions for ethical approval

ACKNOWLEDGEMENT OF CONDITIONS FOR ETHICS APPROVAL

28 March 2024 - 28 March 2025
PRP: Ms M Taljaard
PI: Mr J Page

I, **MS M TALJAARD** (PRP) of the study entitled, **[H24-SCI-CSS-001/U-14] MOBILE ASSESSMENT AND ANNOTATION TOOL** do hereby agree to the following approval conditions:

1. The submission of an annual progress report by myself on the data collection activities of the study by 15 November this year for studies approved in the period October of the previous year up to and including September of this year, or 15 November next year for studies approved after September this year. It is noted that there will be no call for the submission thereof. The onus for submission of the annual report by the stipulated date rests on myself. I am aware of the guidelines (available on Research Ethics Committee (Human) portal) pertinent to the submission of the annual report.

2. Submission of the relevant request to RECH in the event of any amendments to the study for approval by RECH prior to any partial or full implementation thereof. I am aware of the guidelines (available on Research Ethics Committee (Human) portal) pertinent to the requesting for any amendments to the study.

3. Submission of the relevant request to RECH in the event of any extension to the study for approval by RECH prior to the implementation thereof.

4. Immediate submission of a report to RECH in the event of any unanticipated problems, serious incidents or adverse events. I am aware of the guidelines (available on Research Ethics Committee (Human) portal) pertinent to the reporting of any unanticipated problems, serious incidents or adverse events.

5. Immediate discontinuation of the study in the event of any serious unanticipated problems, serious incidents or serious adverse events.

6. Immediate submission of a report to RECH in the event of the unexpected closure/discontinuation of the study (for example, de-registration of the PI).

7. Immediate submission of a report to RECH in the event of study deviations, violations and/or exceptions. I am aware of the guidelines (available on Research Ethics Committee (Human) portal) pertinent to the reporting of any study deviations, violations and/or exceptions.

8. Acknowledgement that the study could be subjected to passive and/or active monitoring without prior notice at the discretion of RECH. I am aware of the guidelines (available on Research Ethics Committee (Human) portal) pertinent to the active monitoring of a study.

Signed: M Taljaard

Date: 28 March 2024

Appendix D: Android Component's External Libraries

ViewLib

ViewLib is a custom library that is found in the RadaeePDF SDK and was utilised for displaying and editing PDFs. This library was essential, as it provided a large part of the functionality of the Android Component, namely question-by-question support, provided the PDF has outlines to extract, and PDF annotation through stylus input. Although this library was very complex and difficult to implement, it provided invaluable in achieving the required functionality for this component, as outlined in Section 3.3.2.

Retrofit2

Retrofit2 is a Gradle-supported library that allows for communication with a Node.js server, as discussed in Section 4.4. Retrofit2 provided a useful and efficient method of asynchronous communication with the server, ensuring that server and client communication occurs seamlessly.

Converter-Gson

Converter-Gson is used in conjunction with Retrofit2, as communication with the server occurs. As the Node.js server sends across JSON files, this library was utilised to parse and decipher data from the JSON file into relevant attributes, encouraging seamless mapping of data from the JSON file to a relevant Java or Kotlin object.

Appendix E: Web Component's External Libraries

JSZip

This library was used to unzip and extract relevant information from zip files containing student submissions. By handling zip files asynchronously, JSZip allowed for the processing of multiple submissions automatically, provided they followed a consistent folder structure, such as the automatic folder structure generated by Moodle when downloading all assessments.

SweetAlert2

SweetAlert2 was integrated into the Web Component to improve the user experience using visually appealing and customisable messages. Using SweetAlert2 was a simple and elegant solution to provide feedback to actions, and indicate the system status where necessary.

Appendix F: Server Component's External Libraries

Express

The Express framework is an open-source web application framework that provides many useful features for facilitating communication between clients and the server. This framework allows you to respond to various communication requests, such as Application Programming Interface (API) calls from clients, creating a simple and powerful centralised communication method.

MySQL2

This package facilitates communication between the server and the MySQL database, which establishes and manages the connection to the database. It allows for efficient data retrieval, editing, and insertion within the MySQL database.

Nodemailer

Nodemailer is used to automatically send emails with attachments, such as PDF or CSV files. It was utilised to achieve the desired functionality of automatically publishing results to students and the moderator.

Child_Process

This package is used to automatically run a separate script, creating child processes automatically. It was used to execute the script responsible for automatically marking submissions by recognising various marking symbols.

BodyParser

This package was used to decipher various incoming requests from clients. All clients send requests via JSON form data. It automatically parses this data, making it accessible within the application.

Multer

This package was used for managing file uploads, particularly for sending emails with attachments and uploading marked submissions to the database. For example, this package was utilised to upload the memorandum, student submission, and the marked student submission to the database.

Appendix G: Marking Symbol Recognition Component's External Libraries

Scikit-learn

Scikit-learn is a machine learning library that was utilised for splitting the dataset into training and validation data. Essentially, this provided a way to monitor the model's progress as it trained, ensuring that the model was able to accurately classify marking symbols.

NumPy

This library offered functionality for arrays and matrices calculations and was utilised for storing and processing image data. The library was also utilised to normalise and ensure consistent dimensions of image data so that the neural network would have consistent data to train from.

PyMuPDF

This toolset provided support for processing PDF files and was used to extract each page as an image, which was then analysed to detect marking symbols. Additionally, this library was used to extract PDF outlines, such as question headings. This allowed for marks to be grouped on a question-by-question basis.

OpenCV

This library was utilised to preprocess the image data such that the data provided to the neural network was standardised.

This ensures that the neural network would be provided standardised data to consistently learn form and accurately classify marking symbols that it was never trained on.

Appendix H: Selection Criteria

NELSON MANDELA UNIVERSITY

WRITTEN INFORMATION PROVIDED TO PARTICIPANT

RESEARCHER'S DETAILS	
Title of the research project	**Mobile Assessment and Annotation Tool**
Reference number	**H24-SCI-CSS-001/U-14**
Principal investigator	**Joshua Page**
Investigator contact telephone number	**061 463 9917**
Investigator email address	**s224046136@mandela.ac.za**

1. Selection process for participants:

1.1	**Selection criteria:**	A representative sample of computer literate students and/or staff within the Department of Computing Sciences would be required. Participants must be over the age of 18 years old, and gender will not be a factor in the selection process.
1.2	**Why are these criteria required:**	You need to be over the age of 18 years old, as well as a student or staff member within the Department of Computing Sciences due to the ethical clearance requirements for this research project.
1.3	**Should you be willing to participate, what is expected of you?**	Participants are expected to meet with the principal investigator at the agreed upon time, date and place to conduct the study. Additionally, participants will be required to complete a post-test questionnaire. If a participant no longer wishes to be included in the study, they may do so with no consequences. If you have any queries, please contact the principal investigator (see contact details above).

2. Relevant risks:

There are no perceived risks associated with this project. If you are a student participant, you may benefit from this evaluation in terms of a learning experience.

Appendix I: Consent Form

CONSENT FORMS

NELSON MANDELA UNIVERSITY

INFORMATION AND INFORMED CONSENT FORM

RESEARCHER'S DETAILS	
Title of the research project	**Mobile Assessment and Annotation Tool**
Reference number	**H24-SCI-CSS-001/U-14**
Principal investigator	**Joshua Page**
Investigator contact telephone number	**061 463 9917**
Investigator email address	**s224046136@mandela.ac.za**

THE FOLLOWING ASPECTS HAVE BEEN EXPLAINED TO ME, THE PARTICIPANT:		
2.1	**Aim:**	The objective is to determine the usability and user experience of the given components of the Mobile Assessment and Annotation Tool.
2.2	**Procedures:**	I understand that I am required to evaluate a system using prescribed tasks. In addition, I will complete a post-test questionnaire.
2.3	**Risks:**	There are no risks attached to this project.
2.4	**Possible benefits:**	There are no benefits attached to this project.
2.5	**Confidentiality:**	My identity will not be revealed in any discussion, description or scientific publications by the investigators.
2.6	**Access to findings:**	Any new information or benefit that develops during the course of the study will be shared on request.
2.7	**Observation:**	Observations may be obtained by the principal investigator during the process of completing tasks.
2.8	**Voluntary participation / refusal / discontinuation:**	My participation is entirely voluntary. My decision whether or not to participate will not affect me in any way.

3.	The information above was explained to me/the participant by Joshua Page in English and I am in command of this language. I was given the opportunity to ask questions and all these questions were answered satisfactorily. No pressure was exerted on me to consent to participation and I understand that I may withdraw at any stage without penalisation. Should I withdraw, all data pertaining to my participation will be destroyed and will no longer form part of the study. Participation in this study will not result in any additional cost to myself.

I HEREBY VOLUNTARILY CONSENT TO PARTICIPATE IN THE ABOVE-MENTIONED PROJECT:	
Signed/confirmed at PORT ELIZABETH on 202_	
Signature of participant:	Signature of witness:
Full name of participant:	Full name of witness:

Appendix J: Usability Test Task List

NELSON MANDELA UNIVERSITY

TASK LIST

RESEARCHER'S DETAILS	
Title of the research project	Mobile Assessment and Annotation Tool
Reference number	H24-SCI-CSS-001/U-14
Principal investigator	Joshua Page
Investigator contact telephone number	061 463 9917
Investigator email address	s224046136@mandela.ac.za

1. Objective of the study

The objective is to determine the usability and user experience of the given components of the Mobile Assessment and Annotation Tool.

2. Main task of the study

The main goal for the user is adding assessments and assessing submission

3. Task list

1. Web-Component (adding assessment)
   1.1. Log in with the following information
      1.1.1. Email: marinda.taljaard@mandela.ac.za
      1.1.2. Password: Password
   1.2. Add a Moodle assessment
   1.3. The assessment must contain the following information:
      - Module: WRRV301
      - Moderator: Isabelle Taljaard
      - Assessment Name: Implementation Document
      - Total Marks: 50
      - Markers:
         - s224046136@mandela.ac.za (Joshua Page)
         - s223037656@mandela.ac.za (Lwando Macakati)
         - s217220207@mandela.ac.za (Dashendri Dorasamy)
      - Assessment memorandum: Navigate to the downloads folder and upload the PDF titled 'Memorandum'
      - Assessment submissions: Navigate to the downloads folder and upload the ZIP file titled 'Moodle Submissions'.

2. Mobile Component
    2.1. Log in with the following information
        2.1.1. Email: s224046136@mandela.ac.za
        2.1.2. Password: Password
            - For the assessment you just added (WRRV301 Implementation Document), download all submissions, by clicking on the three dots for that assessment.
    2.2. When done, select that assessment.
    2.3. Navigate to and then select the submission with student number: 223116831
    2.4. Assess that submission according to the memorandum on the left side of the activity.
    2.5. When done, select the save icon, and set the submission status as marked.
3. Web-Component (Viewing and Publishing Results)
    3.1. Click on the assessment previously added (WRRV301 Implementation Document)
    3.2. Search for the student number: 223116831
    3.3. Click on their submission, and edit their mark to 85.8%
    3.4. Click on 'Publish Results to Students', and then 'Publish Selected'.

Appendix K: Usability Test Questionnaire

Usability Test Questionnaire

Thank for your participating in my usability study. Please complete the following form, selecting whatever option you think is most suitable.

* Indicates required question

1. Please select your participant ID. *

Mark only one oval.

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8

2. I think that I would like to use this system frequently. *

Mark only one oval.

	1	2	3	4	5	
Stro						Strongly Agree

3. I found this system unnecessarily complex. *

Mark only one oval.

	1	2	3	4	5	
Stro						Strongly Agree

4. I thought this system was easy to use. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

5. I think that I would need assistance to be able to use this system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

6. I found the various functions in this system were well integrated. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

7. I thought there was too much inconsistency in this system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agreee

8. I would imagine that most people would learn to use this system very quickly. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agreee

9. I found this system very cumbersome/awkward to use. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

10. I felt very confident using this system. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

11. I needed to learn a lot of things before I could get going with this website. *

Mark only one oval.

	1	2	3	4	5	
Stro	◯	◯	◯	◯	◯	Strongly Agree

12. Please list and briefly explain anything you liked/enjoyed about the system. *

13. Please list and briefly explain anything you disliked/did not enjoy about the system. *